

ME3813

ME3813 Machine Design

Semester 5

Gear box design

By

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1. Table of content

2. Design Parameters

2.1 Given Design Parameters

- Weight of the vehicle = 2050 kg
- No. of passengers = 5
- Speed range = 0 - 170 kmph
- Speed at 30° inclined road = 35 kmph
- No. of forward gears = 5
- No. of reverse gears = 1

Selected Car model: Toyota Highlander



Figure 1 : Toyota Highlander [1]

2.2 Parameters of selected car model

- Maximum speed = 180 kmph [2]
- Drag coefficient = 0.34 [3]
- Width = 1930 mm [2]
- Height = 1755 mm [2]

3. Tyre selection

The tyre used for the Toyota Highlander is 235/55R20 [2]. It has been mentioned in the specification sheet provided by Toyota itself.

Average fully grown adult weighs nearly 90 kg [4]. For safety requirement we use 100kg here.

- Curb Weight = 2050 kg
- Total passenger Weight = 100 kg x 5 = 500 kg
- Total Weight = 2500 + 50 = 2550 kg
- Number of wheels = 4
- Load carried per wheel = 2550/4 = 637.5 kg

3.1 Safety factor calculation for load index and speed rating

Car	TOYOTA Tacoma IV Double Cab	Toyota Prius 2017
Tyre	P245/75R16 109 S	P195/65R15 89 S
Load Index	875	580
Total Weight (kg)	2472	1900
Load per tyre (kg)	618	475
Safety Factor (LI)	1.4159	1.2211
Recommended tyre speed (kmph)	180	180
Max speed (kmph)	160	171
Safety Factor (SR)	1.125	1.053

Table 1 : Safety Factor for load index and speed rating [5]

Calculation of safety factors for load rating and speed index

$$\text{Average Safety Factor for Load Index} = \frac{1.4159 + 1.2211}{2} = 1.3815$$

$$\text{Average Safety Factor for Speed Rating} = \frac{1.125 + 1.053}{2} = 1.089$$

Load per wheel (considering safety factor also) = 637.5 x 1.3815
 = 880.71 kg
 ≈ 900 kg

Speed rating for the wheel = 170 x 1.089
 = 185.13 km/h
 ≈ 190 km/h



Figure 2 : How to read a tyre [6]

3.2 Data related to selected tyre

According load index chart and speed symbol chart we have to select the particular load index and speed symbol

Load Index Chart									
Load index	Load in KG per tyre	Load index	Load in KG per tyre	Load index	Load in KG per tyre	Load index	Load in KG per tyre	Load index	Load in KG per tyre
62	265	75	387	88	560	101	825	114	1180
63	272	76	400	89	580	102	850	115	1215
64	280	77	412	90	600	103	875	116	1250
65	290	78	425	91	615	104	900	117	1285
66	300	79	437	92	630	105	925	118	1320
67	307	80	450	93	650	106	950	119	1360
68	315	81	462	94	670	107	975	120	1400
69	325	82	475	95	690	108	1000	121	1450
70	335	83	487	96	710	109	1030	122	1500
71	345	84	500	97	730	110	1060	123	1550
72	355	85	515	98	750	111	1090	124	1600
73	365	86	530	99	775	112	1120	125	1650
74	375	87	545	100	800	113	1150	126	1700

Table 2 : Tyre load index [7]

Load index for the selected tyre = 104 (for 900 kg)

Speed Symbol Chart		
Speed Rating	Maximum Speed	Type / Vehicle Tyre
L	120 km/h	Off-Road & Light Truck Tyres
M	130 km/h	Temporary Spare Tyre
N	140 km/h	Temporary Spare Tyre
Q	160 km/h	4X4
R	170 km/h	Heavy Duty Light Truck
S	180 km/h	Family Sedans & Vans
T	190 km/h	Family Sedans & Vans
U	200 km/h	Sedans & Coupes
H	210 km/h	Sport Sedans & Coupes
V	240 km/h	Sport Cars
Z	240+ km/h	Sport Cars
W	270 km/h	Exotic Sport Cars
Y	300 km/h	Exotic Sport Cars
(Y)	300+ km/h	Exotic Sport Cars

Table 3 : Speed rating graph [7]

Speed symbol for the selected tyre = T (For 190 km/h)

Nomenclature of selected tyre : **235 / 55 R 20 104 T**

- Width of tyre = 235 mm
- Aspect ratio = 55%
- Side wall height = 126.5 mm
- Rim diameter = 20 inches
- Radius of tyre = side wall height + rim radius
= 126.5 + 254
= 380.5 mm
- Load index of tyre = 104
- Speed symbol = T

4. Force, Power, Torque calculations

4.1 Motion on a Horizontal Road

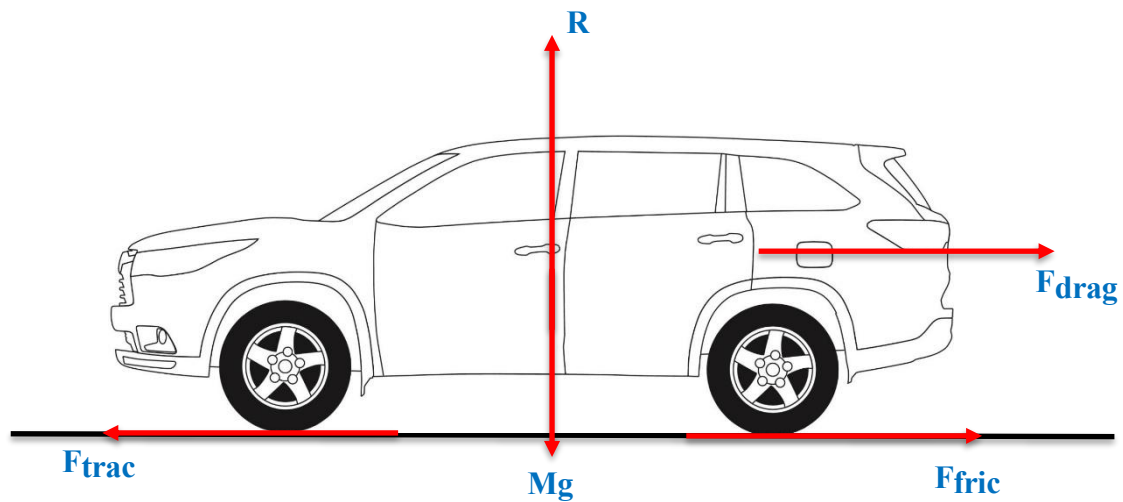


Figure 3: Forces acting on the car at the horizontal motion [8]

4.1.1 Drag force calculation

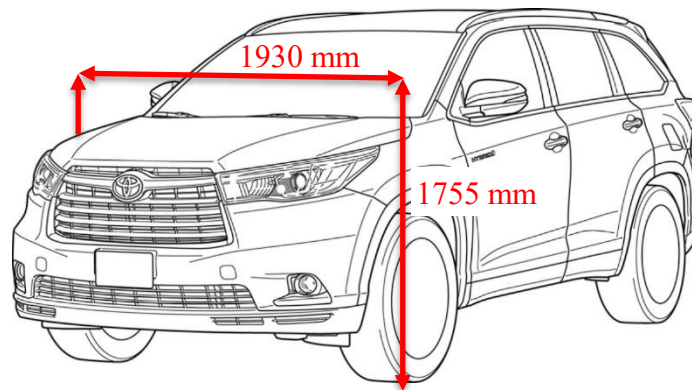


Figure 4 : Frontal Area of the car [9]

- Frontal area = Width x Height
= 1930 mm x 1755 mm
= 3387150 mm²
= 3.38715 m²
- Drag Coefficient (Cd) = 0.34

$$F_{Drag} = \frac{1}{2} \times \rho C_d A V^2$$

Where, ρ = Density of air , V = Speed of vehicle , A = Frontal area , C_d = Drag coefficient

For the worst case:

$$\begin{aligned} \text{Wind speed (V}_{wind}) &= \frac{4.90+4.94+4.37+3.08+4.80+5.30+5.48+5.25+4.68+3.15+3.22+4.70}{12} \\ &= 4.489166667 \text{ ms}^{-1} \\ &\approx 4.5 \text{ ms}^{-1} \end{aligned}$$

After seven years of wind velocity analysis, their average values were obtained and thus the maximum wind velocity was selected. Given below Table 3 are the average values obtained for the monthly wind velocity in those seven years.

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Wind Speed (Km/h)	4.9	4.94	4.37	3.08	4.80	5.30	5.48	5.25	4.68	3.15	3.22	4.70
Average wind direction	N	N	SW	S	SW	SW	SW	SW	SW	SW	NW	N

Table 4 : Average Wind Speed Chart [10]

$$\begin{aligned} \text{Maximum car speed (v)} &= 170 \text{ kmh}^{-1} \\ &= 170 \times \frac{5}{18} \text{ ms}^{-1} \\ &= 47.222222 \text{ ms}^{-1} \\ &\approx 47.5 \text{ ms}^{-1} \end{aligned}$$

$$\begin{aligned} \text{Maximum Relative Velocity of the car} &= \text{Wind speed} + \text{Maximum car speed} \\ &= 4.5 \text{ ms}^{-1} + 47.5 \text{ ms}^{-1} \\ &= 52 \text{ ms}^{-1} \end{aligned}$$

The direction of the wind has been assumed to be parallel to the horizontal plane in this motion on horizontal plane.

$$F_{Drag} = \frac{1}{2} \times \rho C_d A V^2$$

$$F_{Drag} = \frac{1}{2} \times 1.293 \times 0.34 \times 3.38715 \times (52)^2$$

$$F_{Drag} = 2013.20761 \text{ N}$$

4.1.2 Friction force calculation

- Rolling Resistance Coefficient (μ)

$\mu = 0.012$ (According to UAE standard) [11]

$$F_{\text{Friction}} = mg\mu \cos \alpha$$

for horizontal plane $\alpha = 0^\circ$

$$F_{\text{Friction}} = 2550 \times 9.807 \times 0.012 \times \cos(0)$$

$$F_{\text{Friction}} = 300.0942 \text{ N}$$

4.1.3 Slope Resistance Calculation

$$F_{\text{slope Resistance}} = mg \sin(\alpha)$$

for horizontal plane $\alpha = 0^\circ$

$$F_{\text{slope Resistance}} = 2550 \times 9.807 \times \sin(0)$$

$$F_{\text{slope Resistance}} = 0 \text{ N}$$

4.1.4 Traction force calculation

$$F_{\text{traction}} = F_{\text{Drag}} + F_{\text{Friction}} + F_{\text{Slope}}$$

$$F_{\text{traction}} = 2013.20761 + 300.0942 + 0$$

$$F_{\text{traction}} = 2313.30181 \text{ N}$$

4.1.5 Traction power calculation

$$P_{\text{traction}} = F_{\text{traction}} \times V$$

$$P_{\text{traction}} = 2313.30181 \times 47.5$$

$$P_{\text{traction}} = 109881.75 \text{ W}$$

4.1.6 Engine power calculation

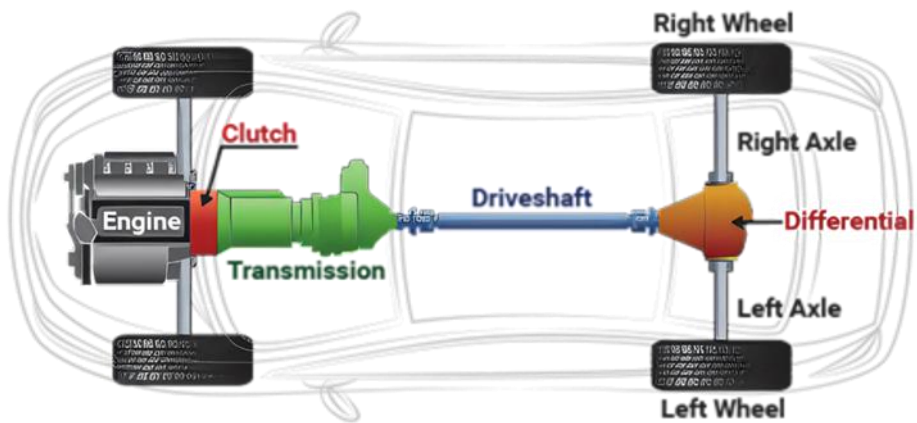


Figure 5 : powertrain diagram plan view [12]

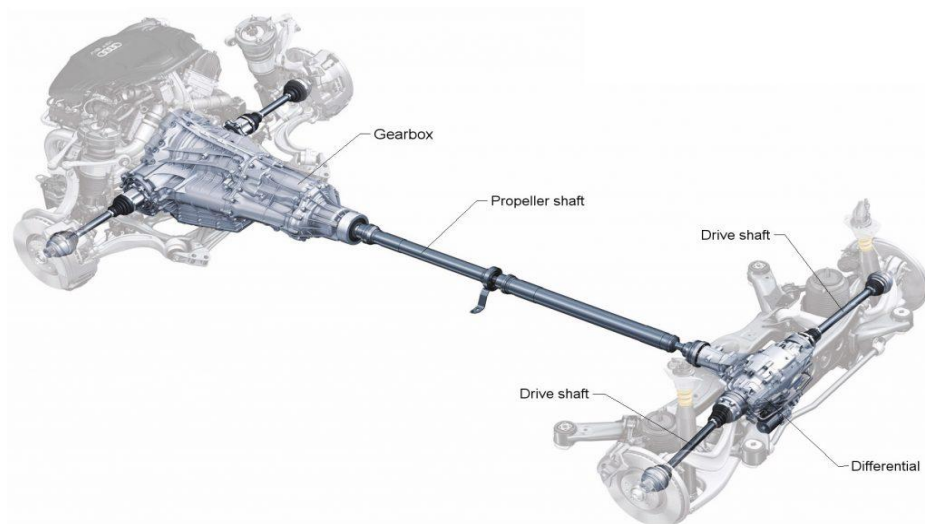


Figure 6 : powertrain diagram 3D view [13]

$\eta_{\text{Drive Shaft}}$	= 98%
$\eta_{\text{Differential}}$	= 93.1%
$\eta_{\text{Gear Box}}$	= 94.1%
$\eta_{\text{Propeller}}$	= 98%
η_{Clutch}	= 85-90 % [14] [15]

$$P_{Engine} = \frac{P_{traction}}{\eta_{Gear\ Box} \times \eta_{Differential} \times \eta_{Drive\ shaft} \times \eta_{Propeller} \times \eta_{Clutch}}$$

$$P_{Engine} = \frac{109881.75}{0.941 \times 0.931 \times 0.98 \times 0.98 \times 0.875}$$

$$P_{Engine} = 148774.625\ W$$

$$P_{Engine} = 148.774625\ kW$$

$$P_{Engine} = 150\ kW$$

4.1.7 Torque calculation

$$T_{wheel} = F_{trac} \times r_{wheel}$$

$$T_{wheel} = 2313.30181 \times 0.38325$$

$$T_{wheel} = 886.5729187\ Nm$$

4.1.8 Angular velocity of wheel calculation

$$\omega_{Wheel} = \frac{V}{r_{wheel}} \times \frac{60}{2\pi}$$

$$\omega_{Wheel} = \frac{47.5}{0.38325} \times \frac{60}{2\pi}$$

$$\omega_{Wheel} = 1183.539694\ rpm$$

$$\omega_{Wheel} = 1184\ rpm$$

4.2 Motion on a 30° Inclined Road

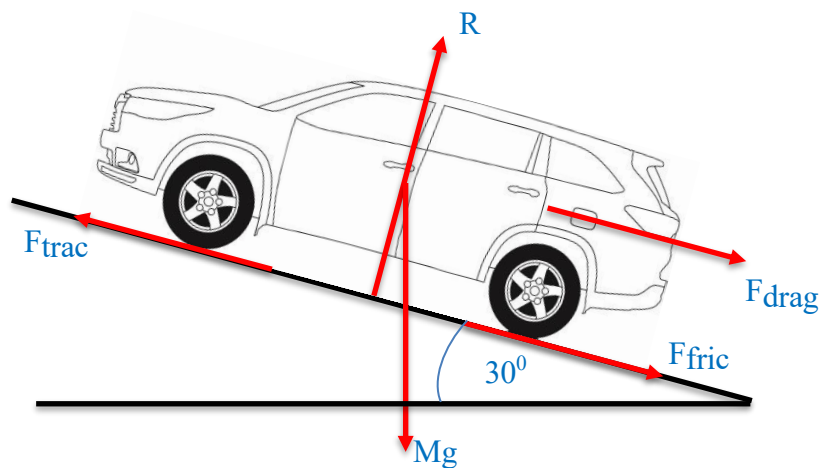


Figure 7: Forces acting on the car at the inclined motion [8]

4.2.1 Drag force calculation

$$\text{Frontal area} = 3.38715 \text{ m}^2$$

$$\text{Drag Coefficient (Cd)} = 0.34$$

$$F_{\text{Drag}} = \frac{1}{2} \times \rho C_d A V^2$$

For the worst case:

$$\text{Wind speed (V}_{\text{wind}}) = 4.5 \text{ ms}^{-1}$$

$$\begin{aligned} \text{Maximum car speed (v)} &= 35 \text{ kmh}^{-1} \\ &= 35 \times \frac{5}{18} \text{ ms}^{-1} \\ &= 9.722222 \text{ ms}^{-1} \\ &\approx 9.75 \text{ ms}^{-1} \end{aligned}$$

$$\begin{aligned} \text{Maximum Relative Velocity of the car} &= \text{Wind speed} + \text{Maximum car speed} \\ &= 4.5 \text{ ms}^{-1} + 9.75 \text{ ms}^{-1} \\ &= 14.25 \text{ ms}^{-1} \end{aligned}$$

The direction of the wind has been assumed to be parallel to the 30° inclined road.

$$F_{\text{Drag}} = \frac{1}{2} \times \rho C_d A V^2$$

$$F_{\text{Drag}} = \frac{1}{2} \times 1.293 \times 0.34 \times 3.38715 \times (14.25)^2$$

$$F_{\text{Drag}} = 151.1860097 \text{ N}$$

4.2.2 Friction force calculation

Rolling Resistance Coefficient (μ)

$$\mu = 0.012$$

$$F_{\text{Friction}} = mg\mu \cos \alpha$$

for inclined plane $\alpha = 30^\circ$

$$F_{\text{Friction}} = 2550 \times 9.807 \times 0.012 \times \cos (30^\circ)$$

$$F_{\text{Friction}} = 259.8892007 \text{ N}$$

4.2.3 Slope Resistance Calculation

$$F_{slope\ Resistance} = mgsin(\alpha)$$

For inclined plane $\alpha = 30^0$

$$F_{slope\ Resistance} = 2550 \times 9.807 \times \sin(30^0)$$

$$F_{slope\ Resistance} = 12503.925\ N$$

4.2.4 Traction force calculation

$$F_{traction} = F_{Drag} + F_{Friction} + F_{Slope}$$

$$F_{traction} = 151.1860097 + 259.8892007 + 12503.925$$

$$F_{traction} = 12915.00021\ N$$

4.2.5 Traction power calculation

$$P_{traction} = F_{traction} \times V$$

$$P_{traction} = 12915.00021 \times 9.75$$

$$P_{traction} = 125921.25\ W$$

4.2.6 Engine power calculation

$$\eta_{Drive\ Shaft} = 98\%$$

$$\eta_{Differential} = 93.1\%$$

$$\eta_{Gear\ Box} = 94.1\%$$

$$\eta_{Propeller} = 98\%$$

$$\eta_{Clutch} = 85 - 90\ \%$$

$$P_{Engine} = \frac{P_{traction}}{\eta_{Gear\ Box} \times \eta_{Differential} \times \eta_{Drive\ shaft} \times \eta_{Propeller} \times \eta_{Clutch}}$$

$$P_{Engine} = \frac{125921.25}{0.941 \times 0.931 \times 0.98 \times 0.98 \times 0.875}$$

$$P_{Engine} = 171040.721\ W$$

$$P_{Engine} = 171.040721\ kW$$

$$P_{Engine} = 172\ kW$$

4.2.7 Torque calculation

$$T_{wheel} = F_{trac} \times r_{wheel}$$

$$T_{wheel} = 12915.00021 \times 0.38325$$

$$T_{wheel} = 4949.67383 \text{ Nm}$$

4.2.8 Angular velocity of wheel calculation

$$\omega_{Wheel} = \frac{V}{r_{wheel}} \times \frac{60}{2\pi}$$

$$\omega_{Wheel} = \frac{9.75}{0.38325} \times \frac{60}{2\pi}$$

$$\omega_{Wheel} = 242.9370951 \text{ rpm}$$

$$\omega_{Wheel} = 243 \text{ rpm}$$

4.3 Summery

Property	At horizontal road	At incident road
Vehicle maximum speed	47.5 ms ⁻¹ (170 kmh ⁻¹)	9.75 ms ⁻¹ (35kmh ⁻¹)
Vehicle maximum speed relative to wind speed	52 ms ⁻¹	14.25 ms ⁻¹
Drag force	2013.278 N	151.186 N
Rolling resistance	300.094 N	259.889 N
Slope resistance	0 N	12503.925 N
Traction force	2313.302 N	12915.000 N
Traction power	109.88 kW	125.92 kW
Engine power	150 kW	172 kW
Wheel torque	886.573 Nm	4949.674 Nm
Wheel maximum angular velocity	1184 rpm	243 rpm

Table 5 : summary of calculation

Maximum power required = 172 kW

5. Engine Selection

5.1 Selected engine

- Selected Engine: DCEC L375-30 Automotive Engine [16]

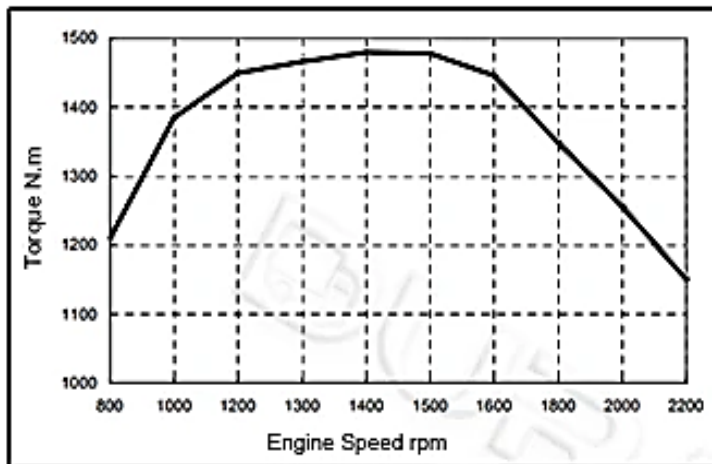


Figure 8 : DCEC L375-30 Automotive Engine [17]

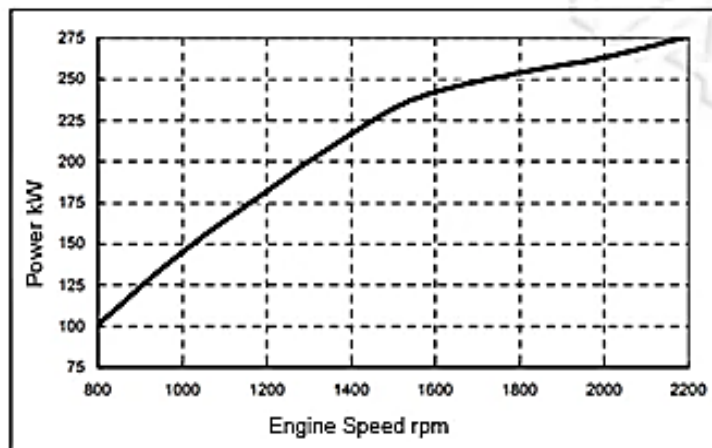
5.2 Prime Mover Parameters

- Engine Model: L375-30
- Engine Type: 4-stroke, mechanical pump
- Displacement: 8.9L
- Rated Speed: 2200 RPM
- Rated Power: 276 kW (375 HP)
- Peak Torque: 1480 Nm at 1400 RPM
- Fuel System: Mechanical pump
- Fuel Type: Diesel
- Cylinders Alignment: Inline 6-cylinder
- Valves per Cylinder: Not specified (but typically 4 for this kind of engine)
- Aspiration Method: Turbocharged and Intercooled
- Emission Standard: Euro III
- Bore × Stroke: Not explicitly mentioned, but usually 114 mm × 145 mm for engines in this class
- Packing Size (Length × Width × Height): 1100 mm × 900 mm × 1250 mm
- Wet Weight: 620 kg

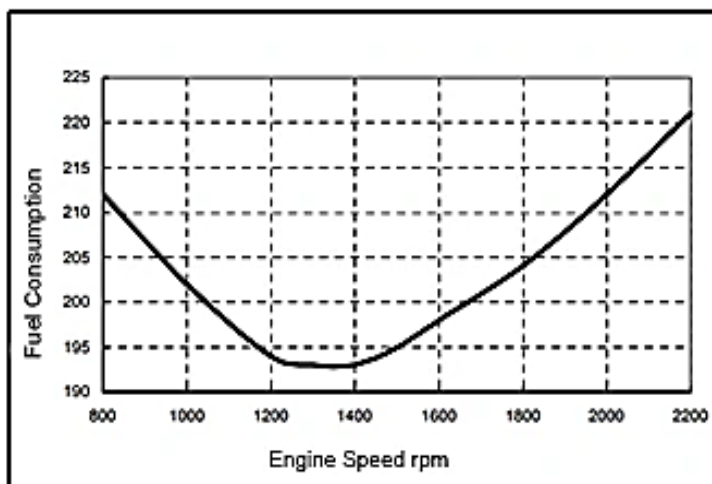
6. Engine characteristic curve



rpm	Torque	N.m
800		1210
1000		1385
1200		1450
1300		1466
1400		1480
1500		1478
1600		1446
1800		1348
2000		1256
2200		1150



rpm	Power	kW
800		101
1000		145
1200		182
1300		200
1400		217
1500		232
1600		242
1800		254
2000		263
2200		276



rpm	Fuel Consumption	g/kW.h
800		212
1000		202
1200		194
1300		193
1400		193
1500		195
1600		198
1800		204
2000		212
2200		221

Graph 1 : Engine characteristic curves [16]

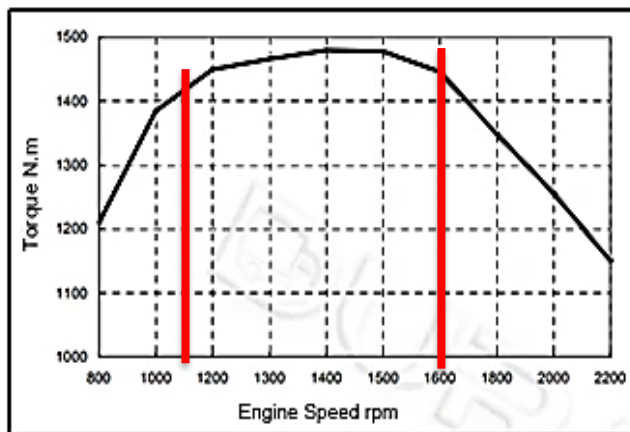
6.1 Optimum RPM region selection

Maximum of fuel consumption is at 2200 rpm. It is 221 g/kWh

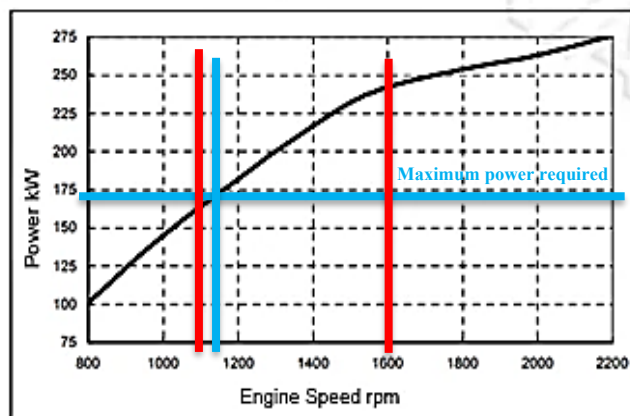
Minimum of fuel consumption is at 1350 rpm. It is 193 g/kWh

So, we can select minimum fuel consumption range (most efficient 20% fuel consumption region) as 193 g/kWh to 198.6 g/kWh which is nearly 1100 to 1600 rpm.

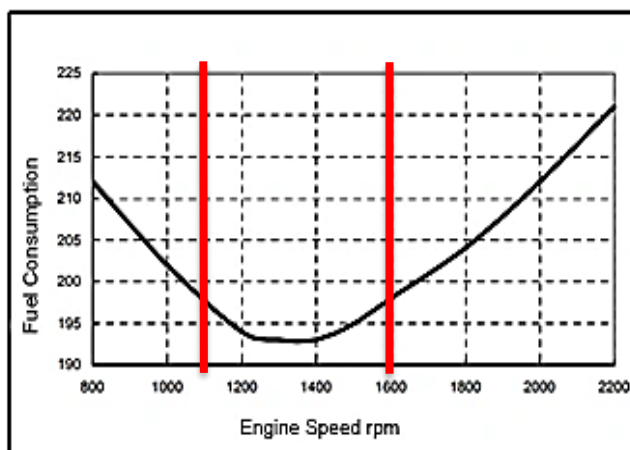
Then selected region = 1100 rpm – 1600 rpm



rpm	Torque	N.m
800		1210
1000		1385
1200		1450
1300		1466
1400		1480
1500		1478
1600		1446
1800		1348
2000		1256
2200		1150



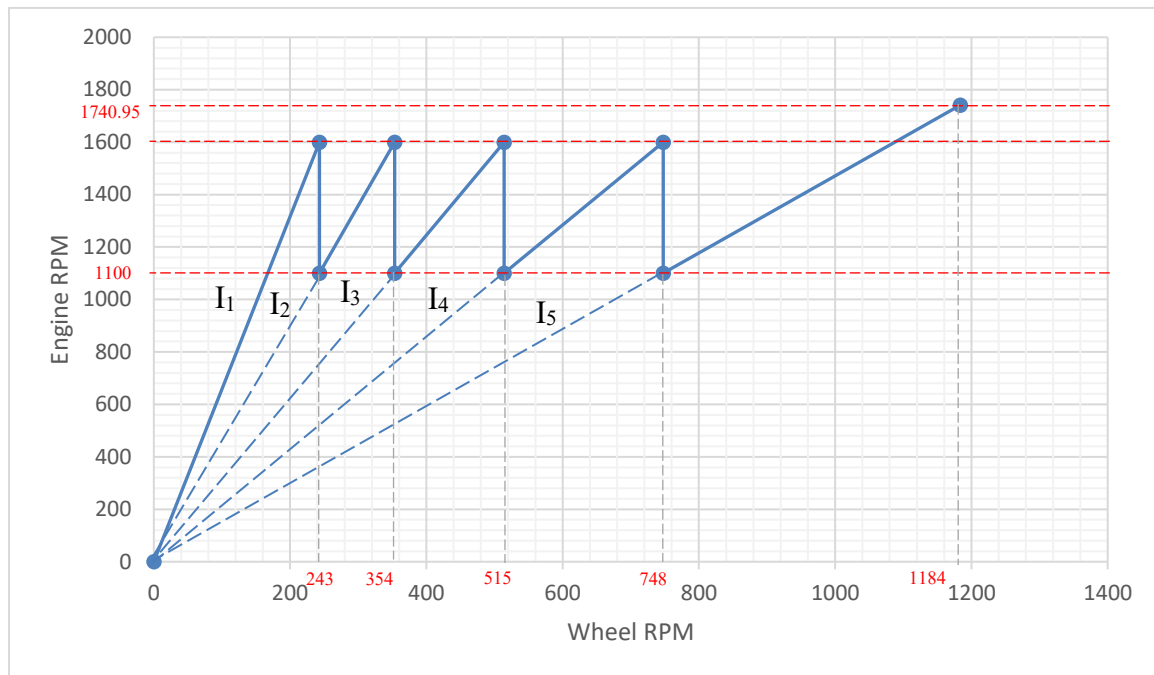
rpm	Power	kW
800		101
1000		145
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1400		217
1500		232
1600		242
1800		254
2000		263
2200		276



rpm	Fuel Consumption	g/kW.h
800		212
1000		202
1200		194
1300		193
1400		193
1500		195
1600		198
1800		204
2000		212
2200		221

Graph 2 : Annotated Version of Engine Characteristic Curve

7. Gear-ratio calculations



Graph 3 : selection of gear ratio

$$r_{wheel} = 0.38325 \text{ m}$$

$$V_1 = 35 \text{ kmh}^{-1} (9.75 \text{ ms}^{-1})$$

$$V_5 = 170 \text{ kmh}^{-1} (47.5 \text{ ms}^{-1})$$

$$L = 1110 \text{ rpm}$$

$$M = 1600 \text{ rpm}$$

7.1 Combined gear ratio calculation

$$\left. \begin{aligned} I_1 &= \frac{M}{\omega_1} \\ I_2 &= \frac{L}{\omega_1} = \frac{M}{\omega_2} \\ I_3 &= \frac{M}{\omega_3} \\ I_4 &= \frac{M}{\omega_4} \\ I_5 &= \frac{M}{\omega_5} \end{aligned} \right\} \quad \frac{I_2}{I_1} = \frac{L}{M} = \frac{1100}{1600} = 0.6875$$

$$\frac{I_2}{I_1} = \frac{I_3}{I_2} = \frac{I_4}{I_3} = \frac{I_5}{I_4} = \frac{L}{M} = 0.6875$$

$$\omega_1 = \frac{V_1}{r_{wheel}} \times \frac{60}{2\pi}$$

$$\omega_1 = \frac{9.75}{0.38325} \times \frac{60}{2\pi}$$

$$\omega_1 = 242.9370951 \text{ rpm}$$

$$\omega_1 = 243 \text{ rpm}$$

$$I_1 = \frac{M}{\omega_1} = \frac{1600}{243} = 6.58436214$$

$$I_2 = \left(\frac{L}{M}\right) \times I_1 = 0.6875 \times 6.58436214 = 4.526748971$$

$$I_3 = \left(\frac{L}{M}\right) \times I_2 = 0.6875 \times 4.526748971 = 3.112139918$$

$$I_4 = \left(\frac{L}{M}\right) \times I_3 = 0.6875 \times 3.112139918 = 2.139596193$$

$$I_5 = \left(\frac{L}{M}\right) \times I_4 = 0.6875 \times 2.139596193 = 1.470972383$$

7.2 Wheel speed calculations

$$\omega_1 = 243 \text{ rpm}$$

$$\omega_2 = \frac{M}{I_2} = \frac{1600}{4.526748971} = 353.4545 \text{ rpm}$$

$$\omega_3 = \frac{M}{I_3} = \frac{1600}{3.112139918} = 514.1157 \text{ rpm}$$

$$\omega_4 = \frac{M}{I_4} = \frac{1600}{2.139596193} = 747.8047 \text{ rpm}$$

$$\omega_5 = \frac{V_5}{r_{wheel}} \times \frac{60}{2\pi}$$

$$= \frac{47.5}{0.38325} \times \frac{60}{2\pi}$$

$$= 1183.53969 \text{ rpm}$$

7.3 Engine RPM calculation at maximum speed

$$R = \omega_5 \times I_5$$

$$R = 1183.53969 \times 1.470972383$$

$$R = 1740.9542 \text{ rpm}$$

$$L < M < R$$

7.4 Gear box gear ratio calculations

Differential gear ratio = 3.605:1

Combined gear ratio(I_n)

= Gear box gear ratio(G_n) $\times$ Differential gear ratio(G_D)

$$I_n = G_n \times G_D$$

$$G_n = \frac{I_n}{G_D}$$

$$G_1 = \frac{I_1}{G_D} = \frac{6.58436214}{3.605} = 1.826452743$$

$$G_2 = \frac{I_2}{G_D} = \frac{4.526748971}{3.605} = 1.255686261$$

$$G_3 = \frac{I_3}{G_D} = \frac{3.112139918}{3.605} = 0.8632843046$$

$$G_4 = \frac{I_4}{G_D} = \frac{2.139596193}{3.605} = 0.5935079592$$

$$G_5 = \frac{I_5}{G_D} = \frac{1.470972383}{3.605} = 0.4080367221$$

Reverse gear ratio = 1st gear ratio = 1.826452743

7.5 Results

Gear	Gear Ratio	Wheel speed in rpm	Wheel speed in kmh ⁻¹
1 st gear	1.826	243	35
2 nd gear	1.256	354	52
3 rd gear	0.863	515	75
4 th gear	0.594	748	109
5 th gear	0.408	1184	170
Reverse gear	1.826	243	35

Table 6 : calculated gear ratios and vehicle speed

The engine rpm is in the optimum region for all gear ratios (without top gear).

8. Gear teeth calculations

Dff

9. Gear teeth calculations

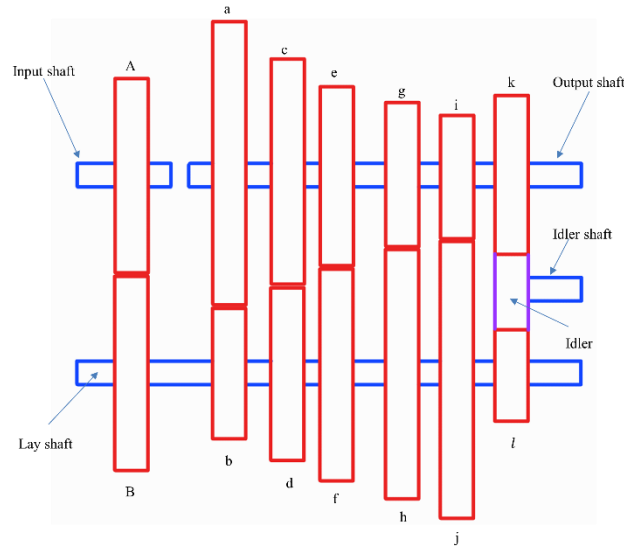


Figure 9 : Diagram of Gear Shafts and Layout for Gear Teeth

According to literature we chose a 20° full depth involute gear system.

S. No.	Systems of gear teeth	Minimum number of teeth on the pinion
1.	$14\frac{1}{2}^\circ$ Composite	12
2.	$14\frac{1}{2}^\circ$ Full depth involute	32
3.	20° Full depth involute	18
4.	20° Stub involute	14

Table 7 : Minimum Number of Teeth on Pinions for Different Gear Teeth Systems

The number of teeth on the pinion (T_P) in order to avoid interference may be obtained from the following relation :

$$T_P = \frac{2A_W}{G \left[\sqrt{1 + \frac{1}{G} \left(\frac{1}{G} + 2 \right) \sin^2 \phi} - 1 \right]}$$

Where,

A_W = Fraction by which the standard addendum for the wheel should be multiplied,

G = Gear ratio or velocity ratio = $T_G/T_P = D_G/D_P$,

ϕ = Pressure angle or angle of obliquity.

First, Input Gear ratio was assumed as 1,

$$\frac{T_A}{T_B} = 1$$

Then minimum allowable distance between shafts were calculated.

$$d = 9.5\sqrt[3]{T_{max}}$$

$$d = 166.7 \approx 168 \text{ mm}$$

Afterwards, minimum number of Gear Teeth can be calculated using Figure 9 and python code mention in appendix,

$$T_A = 42, T_B = 42$$

Based on the Khurmi Gupta's book the minimum number of Gear teeth allowable is 18, and number of Gear Teeth for gears were calculated until 5th Gear.

T_A	42
T_B	42
T_a	54
T_b	30
T_c	47
T_d	37
T_e	39
T_f	45
T_g	31
T_h	53
T_i	24
T_j	60

Table 8 : Calculated Gear Teeth for Each Shaft and Gear in the Transmission System

Reverse Gear

Here the total gear reduction happens in two stages. First stage is between lay shaft and the idler. Teeth on idler gear has been taken as 18 since it do not contribute for gear ratios finally (just to change the direction of rotation).

T_k	33
T_{idler}	18
T_l	18

Table 9 : Calculated Gear Teeth for Reverse Gear Components

After Gear Strength Calculations module is taken as **5 mm**.

8.1 Actual Gear Ratios

Gear	Gear Ratio
1	1.8
2	1.27
3	0.867
4	0.585
5	0.4
Reverse	1.883

Table 10 : Actual Gear Ratios for Each Gear and Reverse

<i>S. No.</i>	<i>Particulars</i>	<i>14½° composite or full depth involute system</i>	<i>20° full depth involute system</i>	<i>20° stub involute system</i>
1.	Addendum	1 <i>m</i>	1 <i>m</i>	0.8 <i>m</i>
2.	Dedendum	1.25 <i>m</i>	1.25 <i>m</i>	1 <i>m</i>
3.	Working depth	2 <i>m</i>	2 <i>m</i>	1.60 <i>m</i>
4.	Minimum total depth	2.25 <i>m</i>	2.25 <i>m</i>	1.80 <i>m</i>
5.	Tooth thickness	1.5708 <i>m</i>	1.5708 <i>m</i>	1.5708 <i>m</i>
6.	Minimum clearance	0.25 <i>m</i>	0.25 <i>m</i>	0.2 <i>m</i>
7.	Fillet radius at root	0.4 <i>m</i>	0.4 <i>m</i>	0.4 <i>m</i>

Table 11 : Gear Tooth Parameters for Different Involute Systems

20° full depth involute system was selected parameters according to chosen module are given below referring Figure 12.

Addendum	5
Dedendum	6.25
Working depth	10
Minimum total depth	11.25
Tooth thickness	7.854
Minimum clearance	1.25
Fillet radius at root	2

Table 12 : selected involute system

Final number of Gear teeth and Pitch Diameters are as below.

Gear	Number of Teeth	Pitch Diameter
A	42	210
B	42	210
a	54	270
b	30	150
c	47	235
d	37	185
e	39	195
f	45	225
g	31	155
h	53	265
i	24	120
j	60	300
k	33	165
l	18	90
idler	18	90

Table 13 : Final number of Gear teeth and Pitch Diameters

8.2 Gear Strength Calculations And Materials Selection For Gears

After several iterations AISI 8620 Steel, oil quenched, 50 mm(2in.) round [845°C(1550°F) quench, 540°C(1000°F) temper] was selected with below properties , and module was selected as 4 **mm**.

Maximum Allowable Tensile Strength	800 MPa
Young's Modulus	205 GPa
Brinell Hardness	900 B.H.N

Table 14 : Mechanical Properties of AISI 8620 Steel Selected for Gear Strength Calculations

8.3 Tangential Load Calculations

$$W_T = \frac{P}{v} \times C_S$$

W_T = Permissible tangential tooth load in newtons

P = Power transmitted in watts

v = Pitch line velocity in m/s = $\frac{\pi DN}{60}$

D = Pitch circle diameter in meters

N = Speed in r.p.m.

C_S = Service factor

Type of load	Type of service		
	Intermittent or 3 hours per day	8-10 hours per day	Continuous 24 hours per day
Steady	0.8	1.00	1.25
Light shock	1.00	1.25	1.54
Medium shock	1.25	1.54	1.80
Heavy shock	1.54	1.80	2.00

Table 15 : Service Factors for Gear Load Based on Type of Load and Duration of Service

We have chosen service factor (C_S) as 1.54 considering medium shock and 8 – 10 hours per day.

Below Table shows the Maximum Engine (Input Shaft) angular velocities for each Gear. Then Lay Shaft, Output Shaft, and Idler Shaft maximum angular velocities were calculated for new actual Gear ratios. So, Speed in rpm was calculated for each gear as below.

Gear	Input Shaft Angular Velocity in Rpm	Input Gear ratio	Lay Shaft Angular Velocity in rpm	Gear ratios	Output Shaft Angular Velocity in rpm
1	1600	1	1600	1.8	888.89
2	1600	1	1600	1.27	1259.84
3	1600	1	1600	0.867	1845.44
4	1600	1	1600	0.585	2735.04
5	1741	1	1741	0.4	4352.5
Reverse	1600	1	1600	1.833	872.89

Table 16 : Calculated Angular Velocities for Input, Lay, and Output Shafts Based on Gear Ratios

$$W_T = (\sigma_o \cdot C_v)b \cdot \pi m \cdot y$$

Where,

W_T = Permissible tangential tooth load or beam strength of gear tooth

σ_o = Allowable static stress

C_v = Velocity factor

b = Face width

m = Module

y = Tooth form factor or Lewis factor

$$y = 0.124 - \frac{0.684}{T}, \text{ for } 14\frac{1}{2}^\circ \text{ composite and full depth involute system.}$$

$$= 0.154 - \frac{0.912}{T}, \text{ for } 20^\circ \text{ full depth involute system.}$$

$$= 0.175 - \frac{0.841}{T}, \text{ for } 20^\circ \text{ stub system.}$$

$$C_v = \frac{3}{3 + v}, \text{ for ordinary cut gears operating at velocities upto } 12.5 \text{ m / s.}$$

$$= \frac{4.5}{4.5 + v}, \text{ for carefully cut gears operating at velocities upto } 12.5 \text{ m/s.}$$

$$= \frac{6}{6 + v}, \text{ for very accurately cut and ground metallic gears operating at velocities upto } 20 \text{ m / s.}$$

$$= \frac{0.75}{0.75 + \sqrt{v}}, \text{ for precision gears cut with high accuracy and operating at velocities upto } 20 \text{ m / s.}$$

$$= \left(\frac{0.75}{1 + v} \right) + 0.25, \text{ for non-metallic gears.}$$

In the above expressions, v is the pitch line velocity in metres per second.

Since we have chosen 20° full depth involute system,

$$y = 0.154 - \frac{0.912}{T}$$

Since gears run below 20 ms^{-1} ,

$$C_v = \frac{6}{6 + v}$$

σ_0 = Allowable static strength should be less than Maximum Allowable Tensile Strength.

This was checked for each gear as below.

Gear Name	Number of Teeth	Pitch Circle Diameter D(mm)	Speed N(rpm)	Pitch Line Velocity v(m/s)	Velocity factor C_v	Lewis factor	Permissible Tangential Tooth Load WT	Allowable Static Stress σ ₀	Maximum Allowable Tensile Strength σ	Test (σ ₀ < σ)
A	42	210	2000	21.9911485	0.2143534	0.13228571	12.04484609	430.3858972	800	SAFE
a	54	270	888.89	12.5663863	0.3231646	0.13711111	21.07845432	481.9949607	800	SAFE
c	47	235	1259.8	15.5017910	0.2790465	0.13459574	17.08705786	460.9561367	800	SAFE
e	39	195	1845.4	18.8422674	0.2415238	0.13061538	14.05775611	451.5043175	800	SAFE
g	31	155	2735.0	22.1969857	0.2127887	0.12458064	11.93315179	456.0960153	800	SAFE
i	24	120	4352.5	27.3475640	0.1799231	0.116	9.685688989	470.2032753	800	SAFE
k	33	165	872.89	7.54122823	0.4430912	0.12636363	35.1242519	635.6119156	800	SAFE
B	42	210	2000	21.9911485	0.2143534	0.13228571	12.04484609	430.3858972	800	SAFE
b	30	150	1600	12.5663706	0.3231649	0.1236	21.07848066	534.6835866	800	SAFE
d	37	185	1600	15.4985237	0.2790889	0.12935135	17.09066	479.6732586	800	SAFE
f	45	225	1600	18.8495559	0.2414530	0.13373333	14.05232044	440.9364602	800	SAFE
h	53	265	1600	22.2005880	0.2127615	0.13679245	11.93121547	415.3648302	800	SAFE
j	60	300	1741	27.3475640	0.1799231	0.1388	9.685688989	392.9652733	800	SAFE
l	18	90	1600	7.53982236	0.4431372	0.10333333	35.13080111	777.3374391	800	SAFE
idler	18	90	1600	7.53982236	0.4431372	0.10333333	35.13080111	777.3374391	800	SAFE

Table 17 : Safety Check of Gears Based on Pitch Diameter, Velocity, and Stress Parameters

8.4 Dynamic Tooth Load Calculations

$$W_D = W_T + W_I = W_T + \frac{21v(b \cdot C + W_T)}{21v + \sqrt{b \cdot C + W_T}}$$

where,

W_D = Total dynamic load in newtons,

W_T = Steady transmitted load in newtons,

v = Pitch line velocity in m/s,

b = Face width of gears in mm , and

C = A deformation or dynamic factor in N/mm.

The value of C in N/mm may be determined by using the following relation :

$$C = \frac{K \cdot e}{\frac{1}{E_P} + \frac{1}{E_G}}$$

Where,

K = A factor depending upon the form of the teeth.

= 0.107, for $14\frac{1}{2}^\circ$ full depth involute system.

= 0.111, for 20° full depth involute system.

= 0.115 for 20° stub system.

E_p = Young's modulus for the material of the pinion in N/mm².

E_G = Young's modulus for the material of gear in N/mm².

e = Tooth error action in mm .

Module (m) in mm	Tooth error in action (e) in mm		
	First class commercial gears	Carefully cut gears	Precision gears
Upto 4	0.051	0.025	0.0125
5	0.055	0.028	0.015
6	0.065	0.032	0.017
7	0.071	0.035	0.0186
8	0.078	0.0386	0.0198
9	0.085	0.042	0.021
10	0.089	0.0445	0.023
12	0.097	0.0487	0.0243
14	0.104	0.052	0.028
16	0.110	0.055	0.030
18	0.114	0.058	0.032
20	0.117	0.059	0.033

Table 18 : Tooth Error in Action for Different Gear Types Based on Module Size

$$K = 0.111, E_p = E_G = 205\text{GPa}, e = 0.015$$

$$\therefore C = 170.663$$

8.5 Static Tooth Load Calculations

$$W_S = \sigma_e \cdot b \cdot p_c \cdot y = \sigma_e \cdot b \cdot \pi m \cdot y$$

σ_e = Flextural endurance = 1.75 × Brinell Hardness (in MPa)

σ_e = 1575 MPa(for 320 B.H.N)

For safety, against tooth breakage, the static tooth load (W_S) should be greater than the dynamic load (W_D). Buckingham suggests the following relationship between W_S and W_D .

For steady loads, $W_S \geq 1.25W_D$

For pulsating loads, $W_S \geq 1.35W_D$

For shock loads, $W_S \geq 1.5W_D$

This relation was checked for each gear as below.

Gear name	Increment Load Due To Dynamic Action WI	WD = WI+WT	Static Tooth Load WS	TEST ($W_s \geq 1.5W_D$)
A	8.767953	20.8127989	205.63	SAFE
a	7.714122	28.7925766	213.13	SAFE
c	8.14665	25.2337078	209.22	SAFE
e	8.508857	22.5666135	203.04	SAFE
g	8.782794	20.7159459	193.66	SAFE
i	9.092511	18.7782002	180.32	SAFE
k	6.500601	41.6248528	196.43	SAFE
B	8.767953	20.8127989	205.63	SAFE
b	7.71412	28.7926003	192.13	SAFE
d	8.146238	25.2368976	201.07	SAFE
f	8.509536	22.5618568	207.88	SAFE
h	8.783052	20.7142673	212.64	SAFE
j	9.092511	18.7782002	215.76	SAFE
l	6.500124	41.6309255	160.63	SAFE
idler	6.500124	41.6309255	160.63	SAFE

Table 19 : Static tooth load calculations

8.6 Wear Tooth Load Calculations

$$W_w = D_p \cdot b \cdot Q \cdot K$$

Where,

W_w = Maximum or limiting load for wear in newtons,

D_p = Pitch circle diameter of the pinion in mm ,

b = Face width of the pinion in mm ,

Q = Ratio factor

$$= \frac{2 \times V \cdot R}{V \cdot R + 1} = \frac{2T_G}{T_G + T_P}, \text{ for external gears}$$

$$= \frac{2 \times V \cdot R}{V \cdot R - 1} = \frac{2T_G}{T_G - T_P}, \text{ for internal gears.}$$

$V \cdot R$ = Velocity ratio = T_G/T_P ,

K = Load-stress factor (also known as material combination factor) in N/mm^2

$$K = \frac{(\sigma_{es})^2 \sin \phi}{1.4} \left(\frac{1}{E_P} + \frac{1}{E_G} \right)$$

Where,

σ_{es} = Surface endurance limit in MPa or N/mm^2

ϕ = Pressure angle

E_P = Young's modulus for the material of the pinion in N/mm^2

E_G = Young's modulus for the material of the gear in N/mm^2

$$\sigma_{es} = \text{Flextural endurance limi} = 2.8 \times \text{Brinell Hardness} - 70$$

$$\sigma_{es} = 2646 \text{ MPa (for 285 B. H. N)}$$

$$\phi = 20^\circ, E_P = E_G = 205 \text{ GPa}$$

$$\therefore K = 16.687$$

The maximum limiting wear load (W_w) must be greater than the dynamic load (W_D). This relation was checked for each gear as below.

	Number of Pinion Teeth	Number of Gear Teeth	Ratio Factor Q	Pinion Diameter Dp	The Maximum Wear Load W_w	TEST ($W_w > W_D$)
Input	42	42	1	210	188.769191	SAFE
1st	30	54	1.2857143	150	173.359461	SAFE
2nd	37	47	1.1190476	111	111.656335	SAFE
3rd	39	45	1.0714286	117	112.68365	SAFE
4th	31	53	1.2619048	93	105.492443	SAFE
5th	24	60	1.4285714	72	92.4583794	SAFE
Reverse	18	33	1.2941176	54	62.8173107	SAFE

Table 20 : Comparison of Maximum Wear Load and Dynamic Load for Each Gear

10. Shaft design calculations

9.1 Material selection

AISI 4340 steel with Ultimate Tensile Strength (σ_{UTS}) of 2400 MPa was selected as the shaft material of the gear box.

According to the A Textbook of Machine Design by Khurumi and Gupta,

- Maximum allowable bending stress (σ_b)

$$\sigma_b = \sigma_{UTS} \times 0.36 = 2400 \times 0.36 = 864 \text{ MPa}$$

- Maximum allowable shear stress (τ)

$$\tau = \sigma_{UTS} \times 0.18 = 2400 \times 0.18 = 432 \text{ MPa}$$

- Combined shock and fatigue factor for bending

$$K_m = 1.5$$

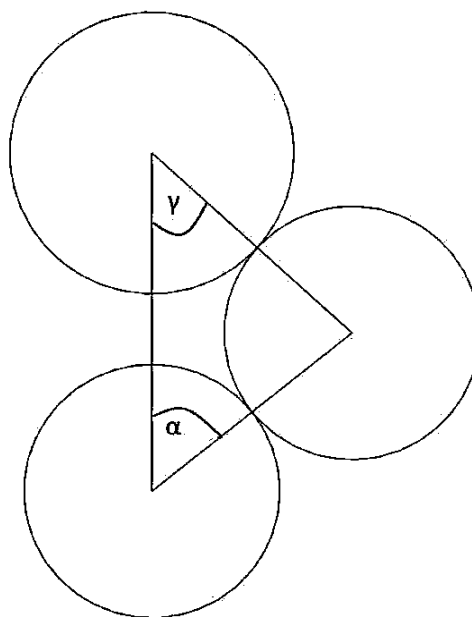
- Combined shock and fatigue factor for torsion

$$K_t = 1.5$$

- Factor of safety

$$F_s = 1.5$$

Idler shaft,



$$\alpha = 38.21^\circ$$

$$\gamma = 21.79^\circ$$

Figure 10 : position of idler shaft

The weights of the gear and the shafts are negligible compared to the radial and tangential forces acting on the gears and reaction forces of bearings.

9.2 Overall procedure of the shaft diameter calculation

$P_{\max}$ = Maximum Power (in W)

$\omega_{\min}$ = Minimum angular velocity of the shaft (in rpm)

Safety factor = C_s (1.5)

Design power = P_o

$$P_o = C_s \times P_{\max}$$

T_t = Maximum torque of the shaft (in Nm)

T = No. of teeth in the gear

m = Module of the gear

F_t = Tangential force on gear (in N)

F_r = Radial force on gear (in N)

ϕ = Pressure angle

$$F_t = \frac{T_t}{\text{radius of the gear}} = \frac{2 \times T_t}{\text{Diameter of the gear}} = \frac{2 \times T_t \times 1000}{m \times T}$$

$$F_r = F_t \times \tan \phi$$

- Bending moment calculations are unique for each shaft.

M = maximum bending moment of the shaft (in Nm)

$$T_e = \sqrt{(k_m \times M)^2 + (k_t \times T_t)^2}$$

$$M_e = \frac{1}{2} \times [T_e + (k_m \times M)]$$

d_1 = shaft diameter under T_e

d_2 = shaft diameter under M_e

$$\left. \begin{aligned} T_e &= \frac{\pi}{16} \times \tau \times d_1^3 \\ M_e &= \frac{\pi}{32} \times \sigma_b \times d_2^3 \end{aligned} \right\} \begin{array}{l} \text{The larger value between } d_1 \text{ and } d_2 \text{ is choose as the} \\ \text{output shaft diameter. That value is rounded to get} \\ \text{a standard shaft diameter.} \end{array}$$

When considering the keyway effect (only input shaft, lay shaft and idler shaft), we use a constant mesh shifting mechanism, so the gear is connected to the output shaft without the use of keys.

e = effect of keyway

$$\tau_{new} = e \times \tau$$

$$\sigma_{b_{new}} = e \times \sigma_b$$

d_3 = shaft diameter under T_e

d_4 = shaft diameter under M_e

$$T_e = \frac{\pi}{16} \times \tau_{new} \times d_3^3$$

$$M_e = \frac{\pi}{32} \times \sigma_{b_{new}} \times d_4^3$$

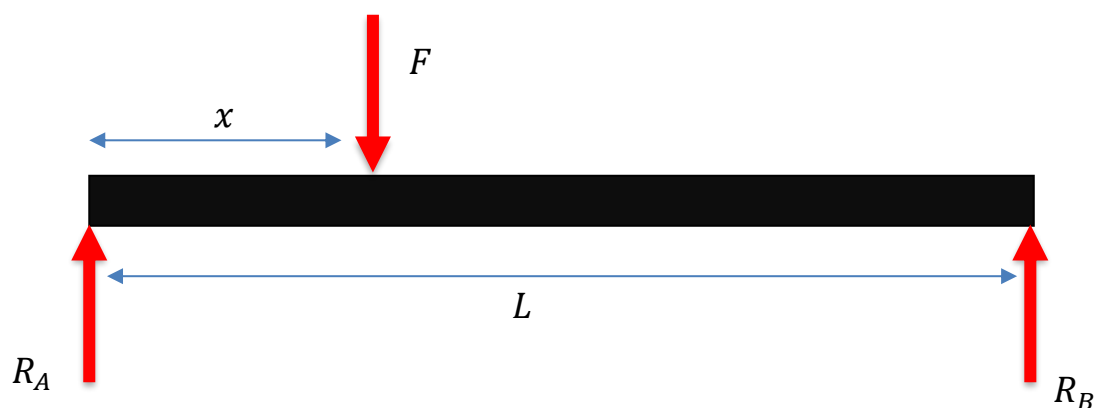
The larger value between d_3 and d_4 is chosen as the diameter (for Input, Idler and Lay shafts). That value is rounded to get a standard shaft diameter.

Diameters (mm)			
5	20	45	90
6	22	50	100
8	25	55	110
10	28	60	120
12	30	65	140
14	32	70	160
16	35	75	180
18	40	80	200

Table 19 : Standard diameters of steel bars used for structural and general engineering purposes

*All the equations were taken from the A Textbook of Machine Design by Khurumi and Gupta.

9.2.1 Bending Moment calculation for Input Shaft, Output Shaft and idler shaft



Take a moment around point A,

$$R_B \times L - F \times x = 0$$

$$R_B = \frac{F \times x}{L}$$

Considering vertical forces on the shaft,

$$R_A + R_B - F = 0$$

$$R_A = F - \frac{F \times x}{L}$$

$$R_A = F \left(1 - \frac{x}{L}\right)$$

Shear Force (S.F.) Diagram

The shear force at any point x can be calculated based on the reactions at the supports and the applied load (in simply supported). The shear force varies along the length of the beam:

Position (x)	S.F.
0	$+R_A$
x	$+R_A - F$
L	0

Table 20 : general shear force values

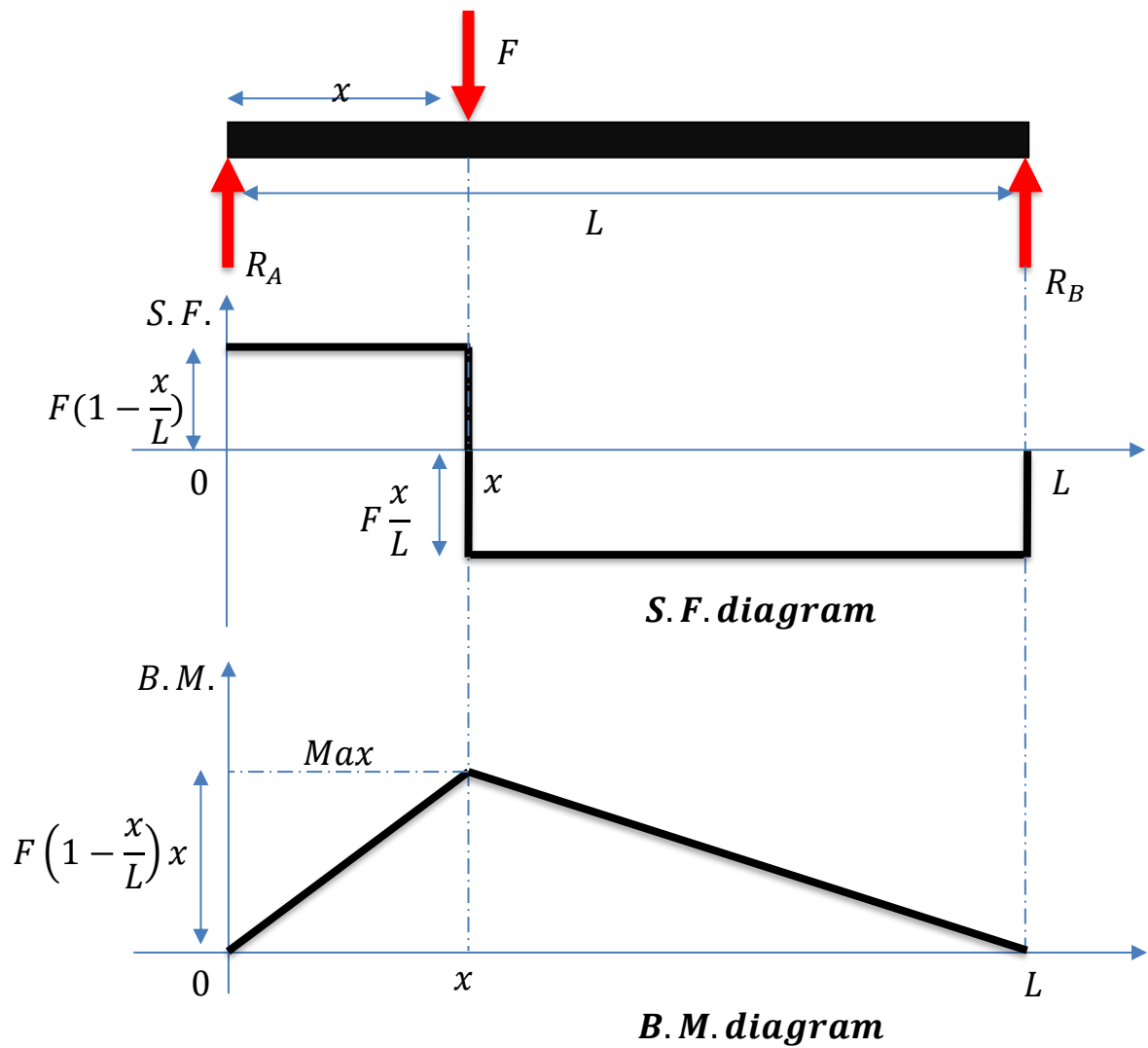
The area under the shear force diagram is equal to the change in the bending moment, which provides insight into how the internal forces are distributed along the beam.

So, according to that,

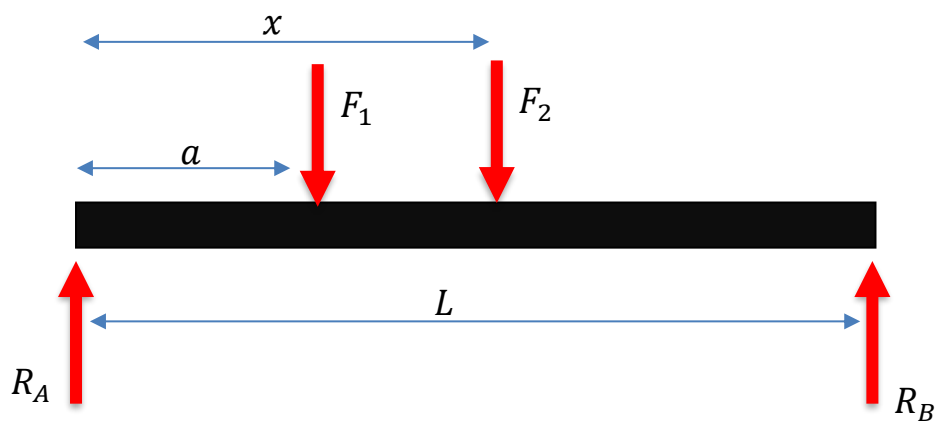
Position (x)	B.M.
0	0
x	$+R_A \times x$
L	0

Table 21 : general bending moment values

Now we can draw S.F. diagram and B.M. diagram according to that.



9.2.2 Bending Moment calculation for lay shaft



Take a moment around point A,

$$R_B \times L - F_1 \times a - F_2 \times x = 0$$

$$R_B = F_2 \times \frac{x}{L} + F_1 \times \frac{a}{L}$$

Considering vertical forces on the shaft,

$$R_A + R_B - F_1 - F_2 = 0$$

$$R_A = F_1 + F_2 - \left(F_2 \times \frac{x}{L} + F_1 \times \frac{a}{L} \right)$$

$$R_A = F_1 \left(1 - \frac{a}{L} \right) + F_2 \left(1 - \frac{x}{L} \right)$$

Shear Force (S.F.) Diagram

The shear force at any point x can be calculated based on the reactions at the supports and the applied load (in simply supported). The shear force varies along the length of the beam:

Position (x)	S.F.
0	$+R_A$
a	$+R_A - F_1$
x	$+R_B$
L	0

Table 22 : general shear force values

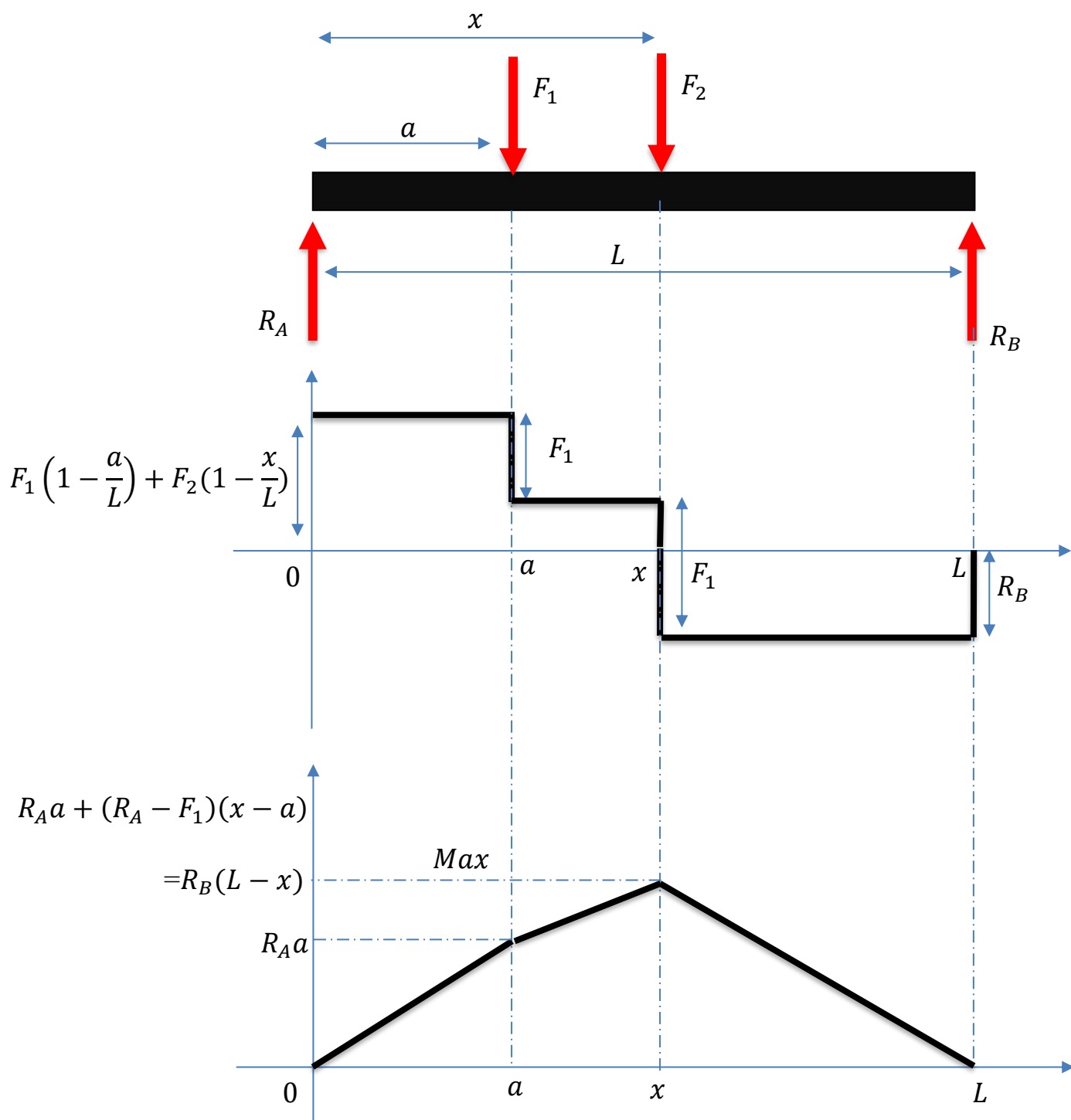
The area under the shear force diagram is equal to the change in the bending moment, which provides insight into how the internal forces are distributed along the beam.

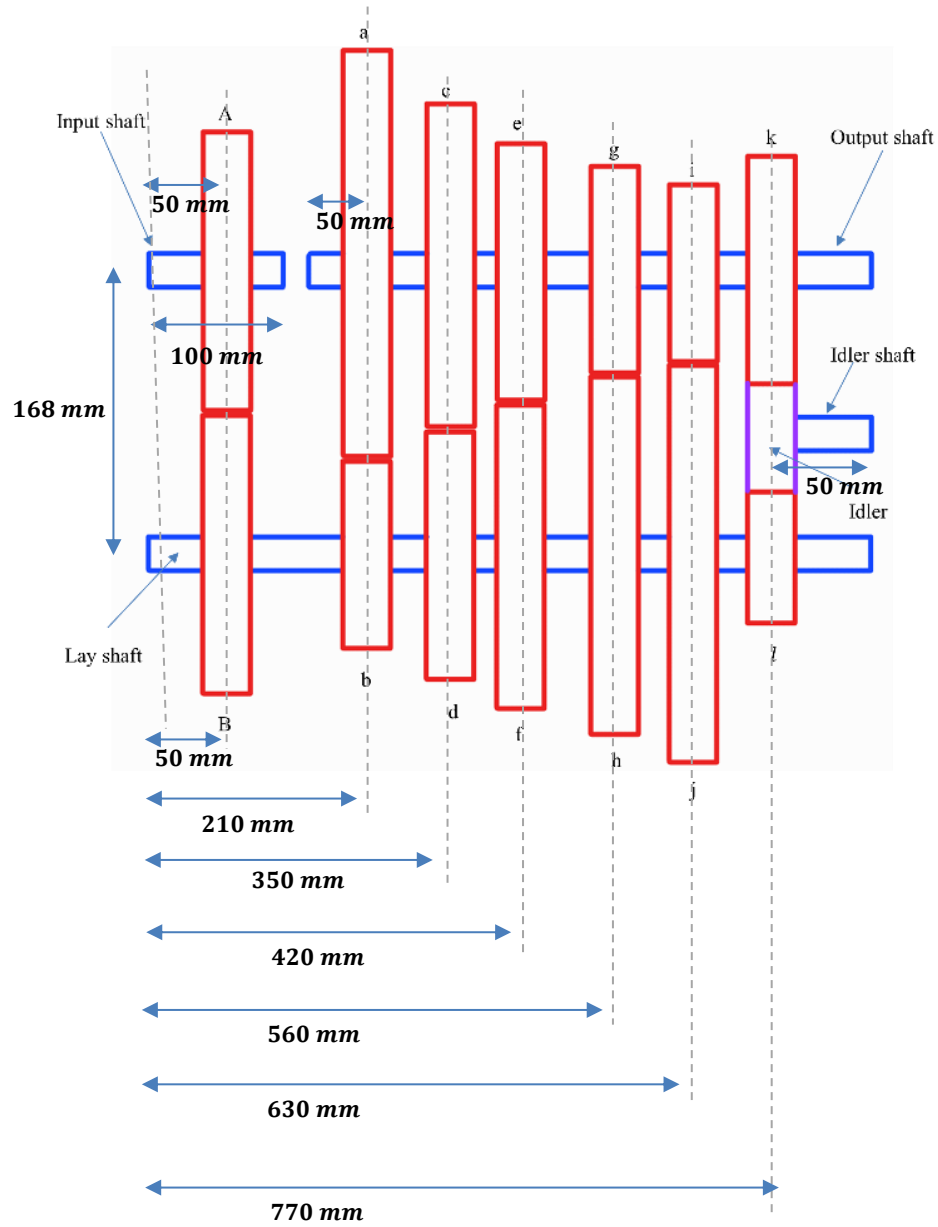
So, according to that,

Position (x)	B.M.
0	0
a	$+R_A \times a$
x	$+R_A \times a + (R_A - F_1) \times (x - a) = +R_B \times (L - x)$
L	0

Table 23 : general bending moment values

Now we can draw S.F. diagram and B.M. diagram according to that.





Shaft	Distance between two Bearings
Input shaft	100 mm
Output shaft	660 mm
Lay shaft	820 mm
Idler shaft	100 mm

Table 24 : distance between bearing

9.3 Input shaft

- Transmitted maximum power = 172 kW
- Transmitted minimum angular velocity = 1400 rpm

Transmitted maximum torque	1759.799 Nm
Tangential force on shaft	16759.99 N
Radial load on the shaft	6098.88 N
Maximum bending moment in tangential direction	418.99 Nm
Maximum bending moment in radial direction	152.47 Nm
Overall Maximum bending moment (M)	445.88 Nm
Equivalent twisting moment (T_e)	2723.1 Nm
Equivalent bending moment (M_e)	1695.96 Nm
Diameter according to T_e	31.78 mm
Diameter according to M_e	27.15 mm
Diameter according to T_e (After considering keyway effect)	34.24 mm
Diameter according to M_e (After considering keyway effect)	29.24 mm
Selected shaft diameter	34.24 mm
Final shaft diameter (after considering keyway effect)	35 mm

Table 25 : input shaft calculation summary

9.3.1 Input shaft calculation

MATERIAL AND PROPERTIES	Steel (AISI 4340)		
	Density	7.85	g/cm ³
	Ultimate tensile strength	864	Mpa
	Modulus of Rigidity	80	Gpa
	Allowable Shear Stress	432	MPa

Input Shaft	Transmitted Power (W) [Pt]	Shaft RPM	Safety Factor [Cs]	Design Power (W) [Pd]	Km	Kt	Transmitted Torque (Nm) [Tt]
Gear A	172000	1400	1.5	258000	1.5	1.5	1759.798942

Tangential force on gear (N) [Ft]	Radial load on the tooth (N) [Fr]	Position of the gear (m)	Maximum bending moment in tangential direction (Mt) [Nm]	Maximum bending moment in radial direction (Mr) [Nm]	Overall Maximum bending moment (M) [Nm]	Equivelant twisting moment (Te) (Nm)
16759.98993	6098.887632	0.05	418.9997481	152.4721908	445.879533	2723.109797

Allowable shear stress (MPa)	Shaft Diameter (mm)	Effect of keyway (e)	Allowable shear stress considering the keyway (MPa)	Shaft Diameter considering the keyway (mm)
432	31.78219139	0.8	345.6	34.23632783

Shaft Diameter_Rounded (mm) Final	Equivelant bending moment (Me) (Nm)	Shaft diameter (mm)	Allowable tensile strength stress considering the keyway (MPa)	Shaft Diameter considering the keyway (mm)	Final Shaft Diameter (mm)
12	1695.964548	27.14153324	691.2	29.23733038	34.23632783
					35

Shaft length (mm)	100		For bearing	
Gear from end(mm)	50		RA	8917.590659
			RB	8917.590659

9.3.2. S.F. and B.M. diagrams of input shaft

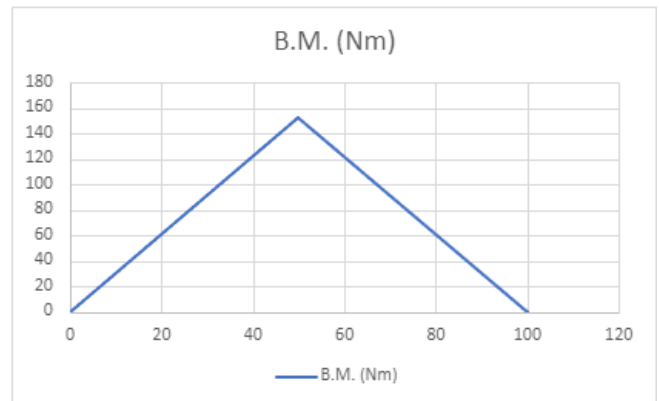
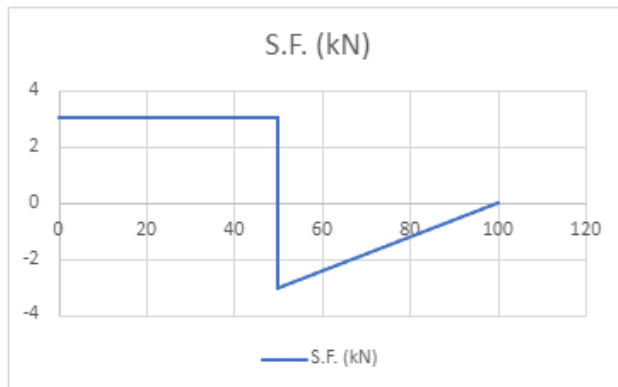
9.3.2.1. For radial loads

Force No.	Forces (kN)	Position (mm)	Moment around A (CC)
1	6.098887632	50	304.9443816

shaft length(mm)	Reaction B (kN)	Reaction A (kN)
100	3.049443816	3.049443816

Position (mm)	S.F. (kN)
0	3.049443816
50	3.049443816
50	-3.049443816
100	0

Position (mm)	B.M. (Nm)
0	0
50	152.4721908
100	0



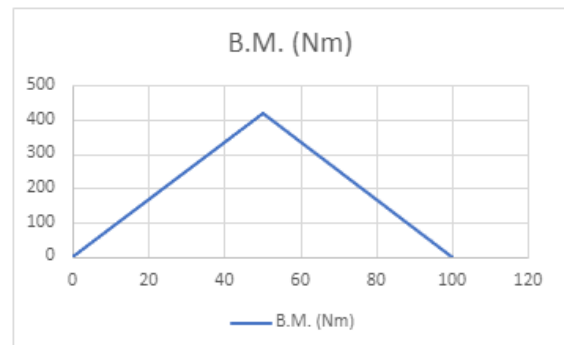
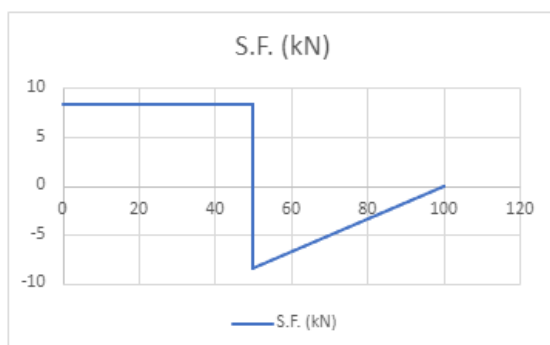
9.3.2.2. For tangential loads

Force No.	Forces (kN)	Position (mm)	Moment around A (CC)
1	39.10664316	50	1955.332158

shaft length(mm)	Reaction B (kN)	Reaction A (kN)
100	19.55332158	19.55332158

Position (mm)	S.F. (kN)
0	19.55332158
50	19.55332158
50	-19.55332158
100	0

Position (mm)	B.M. (Nm)
0	0
50	977.666079
100	0



Shaft diameter (mm)	Under torsion moment		Under bending moment	
	With keyway effect	With keyway effect	Without keyway effect	Without keyway effect
	31.78	34.24	27.14	29.24

∴ Final input shaft diameter = 35 mm

9.4 Idler shaft

Transmitted maximum torque	1759.799 Nm
Tangential force on shaft	39106.64 N
Radial load on the shaft	14230.74 N
Maximum bending moment in tangential direction	977.67 Nm
Maximum bending moment in radial direction	355.77 Nm
Overall Maximum bending moment (M)	1040.39 Nm
Equivalent twisting moment (T_e)	3066.49 Nm
Equivalent bending moment (M_e)	2313.54 Nm
Diameter according to T_e	33.07 mm
Diameter according to M_e	30.10 mm
Diameter according to T_e (After considering keyway effect)	35.62 mm
Diameter according to M_e (After considering keyway effect)	32.43 mm
Selected shaft diameter	35.62 mm
Final shaft diameter (after considering keyway effect)	40 mm

Table 26 : Idler shaft calculation summary

- Transmitted maximum power = 172 kW
- Transmitted minimum angular velocity = 1400 rpm

9.4.1 Idler shaft calculation

	Transmitted Power (W) [Pt]	Shaft RPM	Safety Factor [Cs]	Design Power (W) [Pd]	Km	Kt	Transmitted Torque (Nm) [Tt]
Gear 10	172000	1400	1.5	258000	1.5	1.5	1759.798942
	Maximum RPM	1600					

Tangential force on gear (N) [Ft]	Radial load on the tooth (N) [Fr]	Position of the gear (m)	Maximum bending moment in tangential direction (Mt) [Nm]	Maximum bending moment in radial direction (Mr) [Nm]	Maximum bending moment (M) [Nm]	Equivelant twisting moment (Te) (Nm)	Allowable shear stress (MPa)
39106.64316	14230.73781	0.05	977.666079	355.7684452	1040.385577	3066.498418	432

Shaft Diameter (mm)	Effect of keyway (e)	Allowable shear stress considering the keyway (MPa)	Shaft Diameter considering the keyway (mm)
33.06559708	0.8	345.6	35.6188347

Equivelant bending moment (Me) (Nm)	Shaft diameter (mm)	Allowable tensile strength stress considering the keyway (MPa)	Shaft Diameter considering the keyway (mm)	Final Shaft Diameter (mm)
2313.538392	30.10146906	691.2	32.42582459	35.6188347
				40

Shaft Length (mm) 100
Gear from end(mm) 50

For Bearing	
RA	20807.71154
RB	20807.71154

9.4.2. S.F. and B.M. diagrams of idler shaft

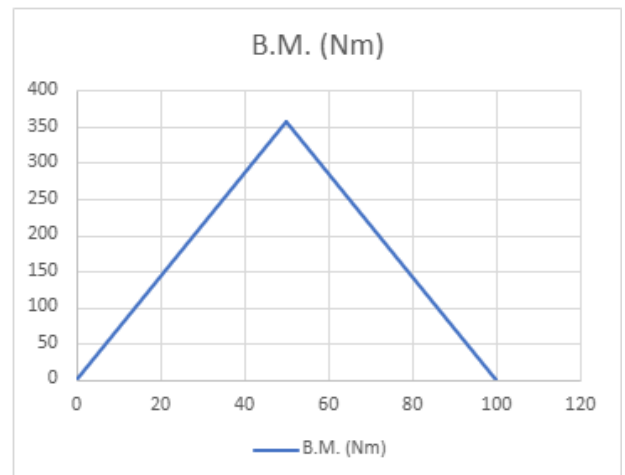
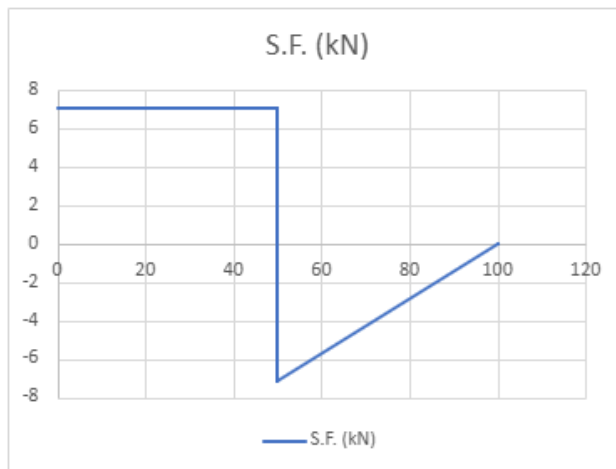
9.4.2.1. For radial loads

Force No.	Forces (kN)	Position (mm)	Moment around A (CC)
1	14.23073781	50	711.5368904

shaft length(mm)	Reaction B (kN)	Reaction A (kN)
100	7.115368904	7.115368904

Position (mm)	S.F. (kN)
0	7.115368904
50	7.115368904
50	-7.115368904
100	0

Position (mm)	B.M. (Nm)
0	0
50	355.7684452
100	0



9.4.2.2. For tangential loads

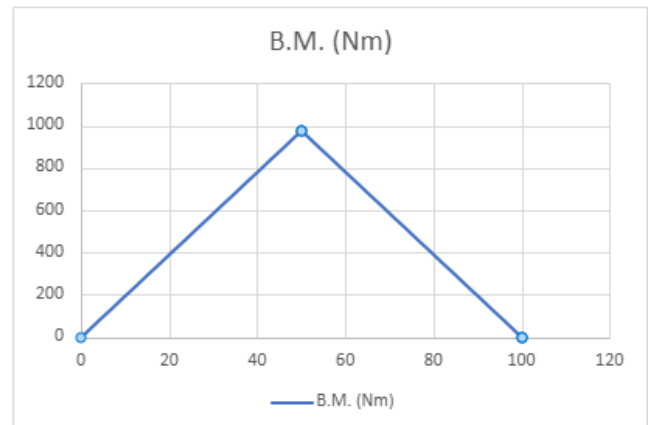
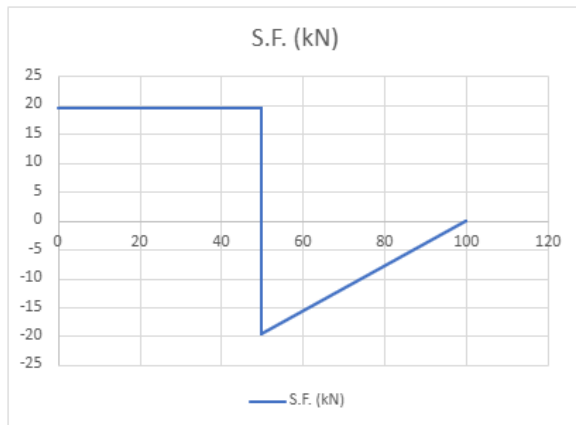
Force No.	Forces (kN)	Position (mm)	Moment around A (CC)
1	39.10664316	50	1955.332158

shaft

length(mm)	Reaction B (kN)	Reaction A (kN)
100	19.55332158	19.55332158

Position (mm)	S.F. (kN)
0	19.55332158
50	19.55332158
50	-19.55332158
100	0

Position (mm)	B.M. (Nm)
0	0
50	977.666079
100	0



Shaft diameter (mm)	Under torsion moment		Under bending moment	
	With keyway effect	With keyway effect	Without keyway effect	Without keyway effect
	33.07	35.62	30.10	32.43

∴ Final idler shaft diameter = 40 mm

9.5 Lay shaft

- Transmitted maximum power = 172 kW
- Transmitted minimum angular velocity = 1400 rpm
- Transmitted maximum torque = 1759.799 Nm
- **Final shaft diameter (after considering keyway effect) = 50 mm**

9.5.1 lay shaft calculation

	Transmitted Power (W) [Pt]	Shaft RPM	Safety Factor [Cs]	Design Power (W) [Pd]	Km	Kt	Transmitted Torque (Nm) [Tt]	
Gear B	172000	1400	1.5	258000	1.5	1.5	1759.798942	
Gear b	172000	1400	1.5	258000	1.5	1.5	1759.798942	
Gear d	172000	1400	1.5	258000	1.5	1.5	1759.798942	
Gear f	172000	1400	1.5	258000	1.5	1.5	1759.798942	
Gear h	172000	1400	1.5	258000	1.5	1.5	1759.798942	
Gear j	172000	1400	1.5	258000	1.5	1.5	1759.798942	
Gear l	172000	1400	1.5	258000	1.5	1.5	1759.798942	
End bearing RB								
	Maximum RPM	1740.95						
Tangential force on gear (N) [Ft]	Radial load on the tooth (N) [Fr]	Position of the gear from left side (m)	RB for Tangential force (N)	RA for Tangential force (N)	RB for Radial force (N)	RA for Radial force (N)	RA Final	RB Final
16759.98993	6098.887632	0.05						
23463.9859	8538.442685	0.21	7031.020164	33192.95566	2558.557738	12078.77258	35322.3591	35414.90181
19024.85343	6923.061636	0.35	9142.314874	26642.52848	3326.848725	9695.100543	28351.70715	28546.22954
15642.65726	5692.295123	0.42	9034.04335	23368.60384	3287.449187	8503.733568	24867.75281	25084.10756
13281.50145	4833.080765	0.56	10092.24428	19949.2471	3672.523915	7259.444482	21229.03658	21544.35949
11731.99295	4269.221342	0.63	10035.55494	18456.42793	3651.894911	6716.214063	19640.44967	19977.07686
39106.64316	14230.73781	0.77	37744.04235	18122.59073	13734.89328	6594.732155	19285.19606	23676.27675

1st Gear

1st									
Position from Left	0	0.05	0.21	0.35	0.42	0.56	0.63	0.77	0.82
Mt (Nm)		1659.647783	4288.9223						
Mr (Nm)		603.9386289	1560.72022						
Total M		1766.117955	4564.0664						
Me	7091.73783	M 6846.0996							
Te	7337.37606								
e	0.8								
tensile stress (Mpa)	691.2	864							
shear stress (Mpa)	345.6	432							
Diameter of the shaft (mm)	Under Me	Under Te	Me (WO keyway)	Te (WO keyway)					
	47.10312087	47.64080064	43.72666397	44.22580165					

2nd gear

2nd									
Position from Left	0	0.05	0.21	0.35	0.42	0.56	0.63	0.77	0.82
Mt (Nm)		1332.126424		4296.887991					
Mr (Nm)		484.7550272		1563.618901					
Total M		1417.585358		4572.543108					
Me	7104.027933	M 6858.814661							
Te	7349.241204								
e	0.8								
tensile stress (Mpa)	691.2	864							
shear sress (Mpa)	345.6	432							
Diameter of the shaft (mm)	Under Me	Under Te							
	47.13031534	47.66646651	43.75190908	44.24962774					

M 6858.814661

3rd gear

3rd									
Position from Left	0	0.05	0.21	0.35	0.42	0.56	0.63	0.77	0.82
Mt (Nm)		1168.430192			3613.61734				
Mr (Nm)		425.1866784			1314.979675				
Total M		1243.387641			3845.439094				
Me	6055.816298	M5768.158641							
Te	6343.473955								
e	0.8								
tensile stress (Mpa)	691.2	864							
shear sress (Mpa)	345.6	432							
Diameter of the shaft (mm)	Under Me	Under Te							
	44.68787356	44.68787356	41.48454698	41.48454698					

M 5768.158641

4th gear

4th									
Position from Left	0	0.05	0.21	0.35	0.42	0.56	0.63	0.77	0.82
Mt (Nm)		997.4623549				2623.983512			
Mr (Nm)		362.9722241				954.8562178			
Total M		1061.451829				2792.31801			
Me	4569.685307	M 4188.477015							
Te	4950.893598								
e	0.8								
tensile stress (Mpa)	691.2	864							
shear sress (Mpa)	345.6	432							

M 4188.477015

Diameter of the shaft (mm)	Under Me	Under Te	Me (WO keyway)	Te (WO keyway)
	40.68436636	41.78560132	37.76802012	38.7903161

5th gear

5th									
Position from Left	0	0.05	0.21	0.35	0.42	0.56	0.63	0.77	0.82
Mt (Nm)		922.8213965					1906.755439		
Mr (Nm)		335.8107031					693.8600331		
Total M		982.0224836					2029.078128		
Me	3536.232417								
Te	4028.847643								
e	0.8								
tensile stress (Mpa)	691.2	864							
shear stress (Mpa)	345.6	432							
Diameter of the shaft (mm)	Under Me	Under Te	Me (WO keyway)	Te (WO keyway)					
	37.3518737	39.01147821	34.67440798	36.21504833					

M 3043.617192

Reverse gear

Reverse									
Position from Left	0	0.05	0.21	0.35	0.42	0.56	0.63	0.77	0.82
Mt (Nm)		906.1295366						1887.202118	
Mr (Nm)		329.7366077						686.7446642	
Total M		964.259803						2008.270417	
Me	3508.863253								
Te	4005.320881								
e	0.8								
tensile stress (Mpa)	691.2	864							
shear stress (Mpa)	345.6	432							
Diameter of the shaft (mm)	Under Me	Under Te	Me (WO keyway)	Te (WO keyway)					
	37.25526067	38.93539309	34.58472039	36.14441716					

M 3012.405625

For Bearing	
RA	35322.3591
RB	35414.90181

Final diameters of the lay shaft for each gear with keyway effect.

Gear	1	2	3	4	5	Reverse
Diameter of the Shaft (mm)	47.64080064	47.66646651	44.68787356	41.78560132	39.01147821	38.93539309
Final diameter of the shaft (mm)	47.66646651					
	50					

9.5.2. S.F. and B.M. diagrams of lay shaft

9.5.2.1. First gear

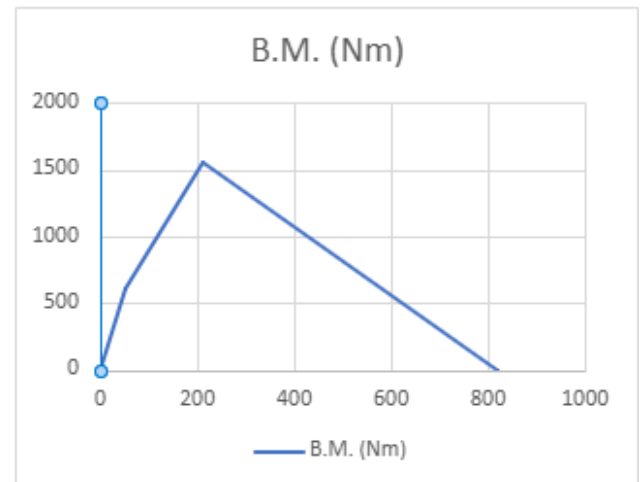
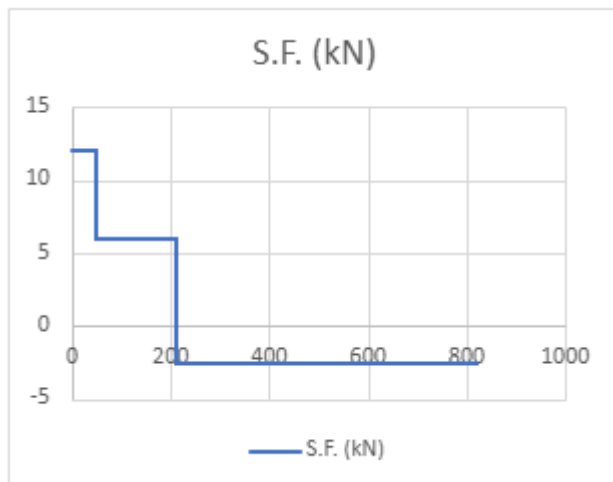
9.5.2.1.1 For radial loads

Force No.	Forces (kN)	Position (mm)	Moment around A (CC)
1	6.098887632	50	304.9443816
2	8.538442685	210	1793.072964

shaft length(mm)	Reaction B (kN)	Reaction A (kN)
820	2.558557738	12.07877258

Position (mm)	S.F. (kN)
0	12.07877258
50	12.07877258
50	5.979884946
210	5.979884946
210	-2.558557738
820	-2.558557738

Position (mm)	B.M. (Nm)
0	0
50	603.9386289
210	1560.72022
820	0



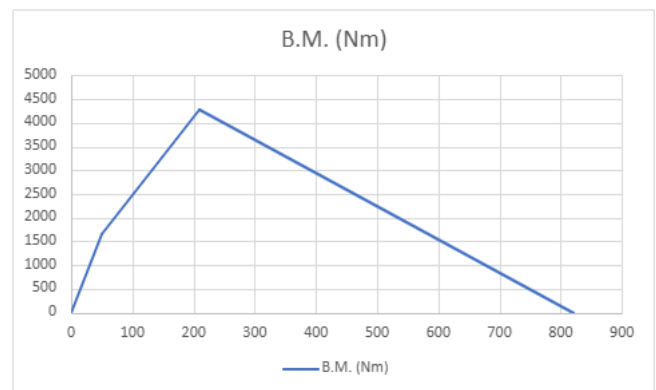
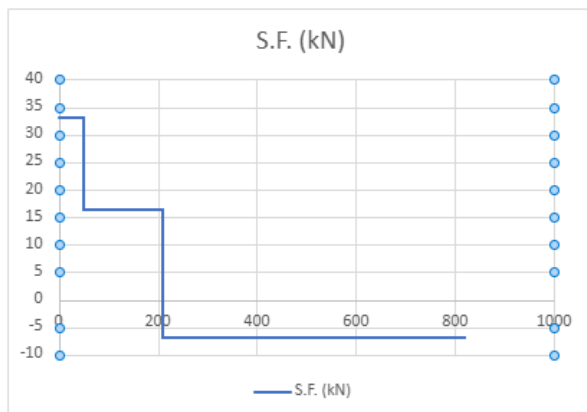
9.5.2.1.2 For tangential loads

Force No.	Forces (kN)	Position (mm)	Moment around A (CC)
1	16.75998993	50	837.9994963
2	23.4639859	210	4927.437038

shaft length(mm)	Reaction B (kN)	Reaction A (kN)
820	7.031020164	33.19295566

Position (mm)	S.F. (kN)
0	33.19295566
50	33.19295566
50	16.43296573
210	16.43296573
210	-7.031020164
820	-7.031020164

Position (mm)	B.M. (Nm)
0	0
50	1659.647783
210	4288.9223
820	0



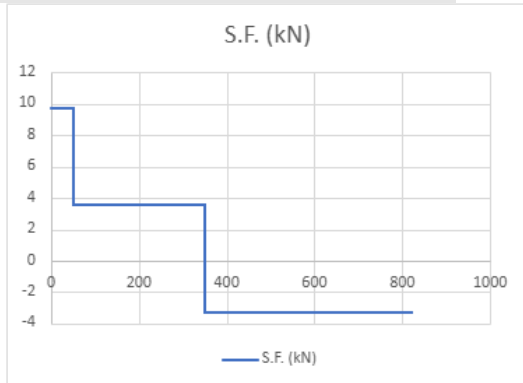
9.5.2.2. Second gear

9.5.2.2.1. For radial loads

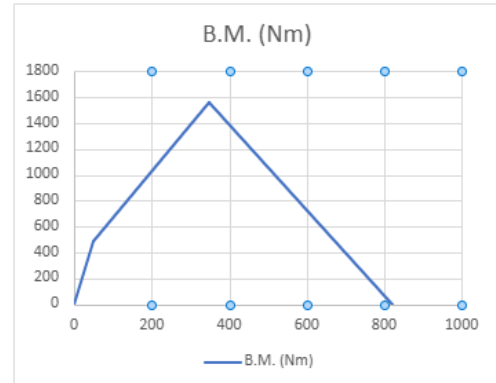
Force No.	Forces (kN)	Position (mm)	Moment around A (CC)
1	6.098887632	50	304.9443816
2	6.923061636	350	2423.071573

shaft length(mm)	Reaction B (kN)	Reaction A (kN)
820	3.326848725	9.695100543

Position (mm)	S.F. (kN)
0	9.695100543
50	9.695100543
50	3.596212911
350	3.596212911
350	-3.326848725
820	-3.326848725



Position (mm)	B.M. (Nm)
0	0
50	484.7550272
350	1563.618901
820	0



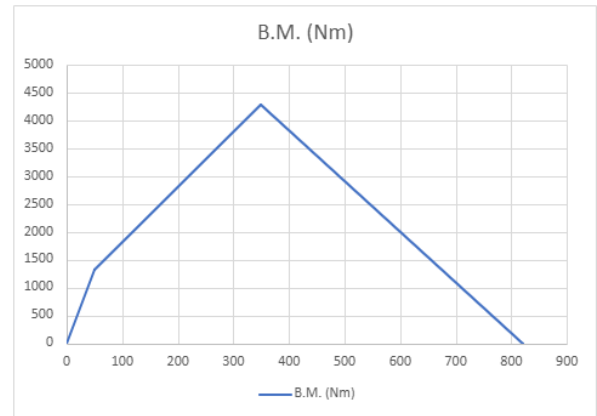
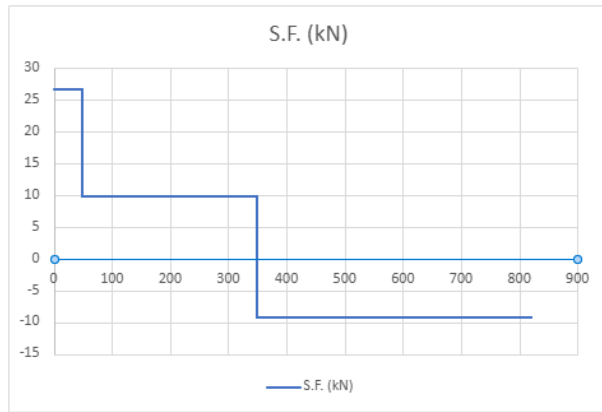
9.5.2.2.2 For tangential loads

Force No.	Forces (kN)	Position (mm)	Moment around A (CC)
1	16.75998993	50	837.9994963
2	19.02485343	350	6658.6987

shaft length(mm)	Reaction B (kN)	Reaction A (kN)
820	9.142314874	26.64252848

Position (mm)	S.F. (kN)
0	26.64252848
50	26.64252848
50	9.882538555
350	9.882538555
350	-9.142314874
820	-9.142314874

Position (mm)	B.M. (Nm)
0	0
50	1332.126424
350	4296.887991
820	0



9.5.2.3. Third gear

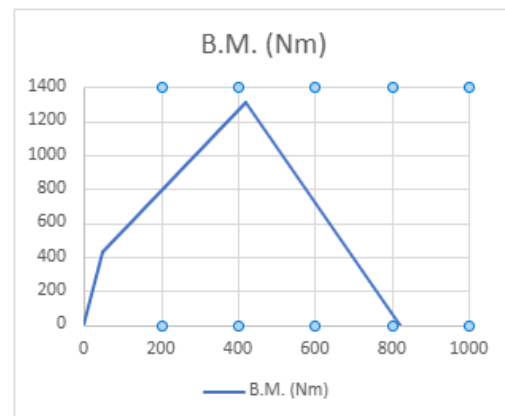
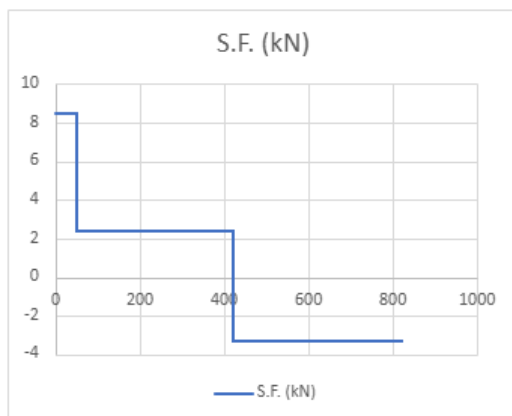
9.5.2.3.1. For radial loads

Force No.	Forces (kN)	Position (mm)	Moment around A (CC)
1	6.098887632	50	304.9443816
2	5.692295123	420	2390.763952

shaft length(mm)	Reaction B (kN)	Reaction A (kN)
820	3.287449187	8.503733568

Position (mm)	S.F. (kN)
0	8.503733568
50	8.503733568
50	2.404845936
420	2.404845936
420	-3.287449187

Position (mm)	B.M. (Nm)
0	0
50	425.1866784
420	1314.979675
820	0



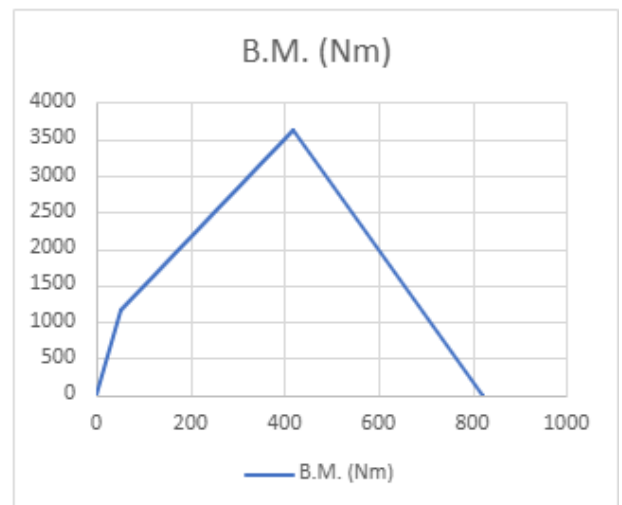
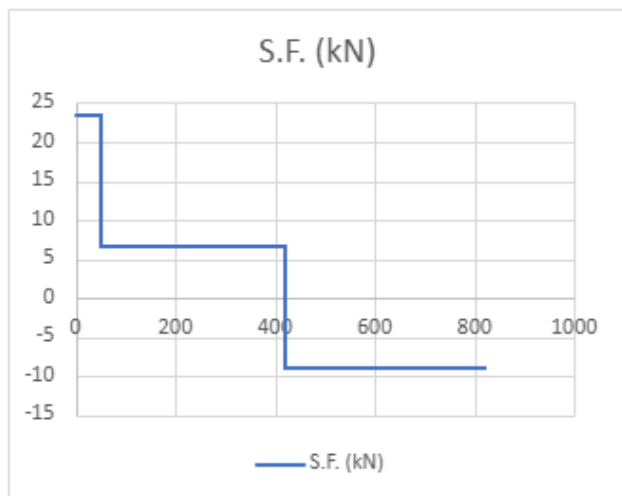
9.5.2.3.2 For tangential loads

Force No.	Forces (kN)	Position (mm)	Moment around A (CC)
1	16.75998993	50	837.9994963
2	15.64265726	420	6569.916051

shaft length(mm)	Reaction B (kN)	Reaction A (kN)
820	9.03404335	23.36860384

Position (mm)	S.F. (kN)
0	23.36860384
50	23.36860384
50	6.608613914
420	6.608613914
420	-9.03404335
820	-9.03404335

Position (mm)	B.M. (Nm)
0	0
50	1168.430192
420	3613.61734
820	0



9.5.2.4. Fourth gear

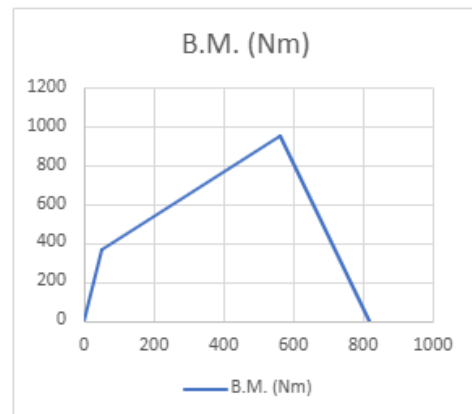
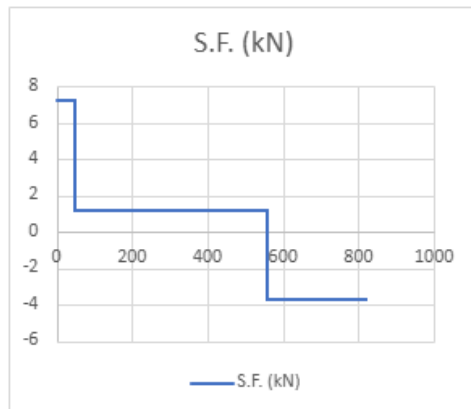
9.5.2.4.1. For radial loads

Force No.	Forces (kN)	Position (mm)	Moment around A (CC)
1	6.098887632	50	304.9443816
2	4.833080765	560	2706.525228

shaft length(mm)	Reaction B (kN)	Reaction A (kN)
820	3.672523915	7.259444482

Position (mm)	S.F. (kN)
0	7.259444482
50	7.259444482
50	1.16055685
560	1.16055685
560	-3.672523915
820	-3.672523915

Position (mm)	B.M. (Nm)
0	0
50	362.9722241
560	954.8562178
820	0



9.5.2.4.2 For tangential loads

Force No.	Forces (kN)	Position (mm)	Moment around A (CC)
1	16.75998993	50	837.9994963
2	13.28150145	560	7437.640812

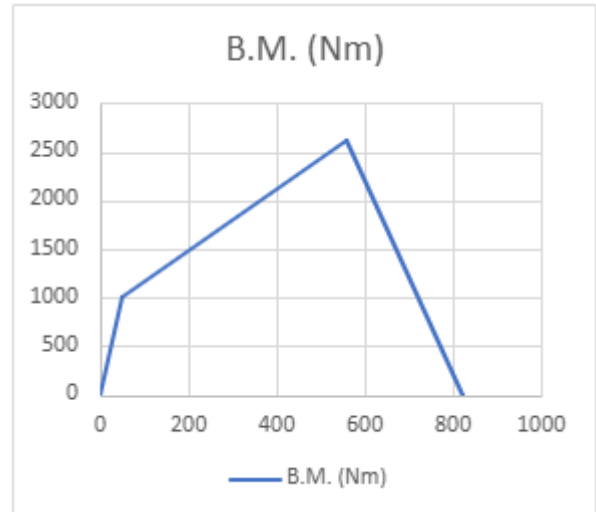
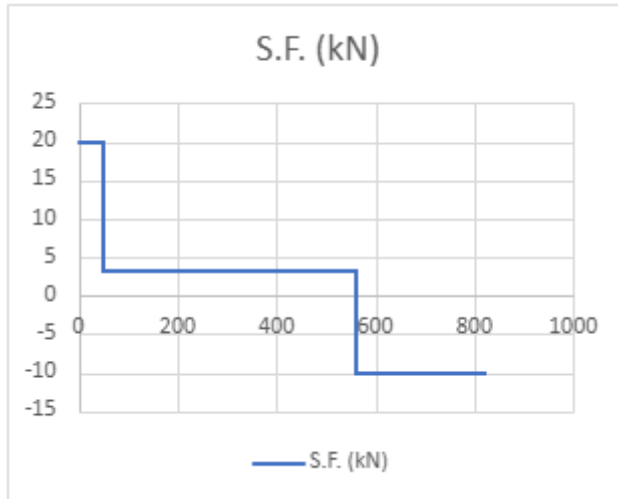
shaft length(mm)	Reaction B (kN)	Reaction A (kN)
820	10.09224428	19.9492471

Position (mm)	S.F. (kN)
0	19.9492471

Position (mm)	B.M. (Nm)
0	0

50	19.9492471
50	3.189257172
560	3.189257172
560	-10.09224428
820	-10.09224428

50	997.4623549
560	2623.983512
820	0



9.5.2.5. Fifth gear

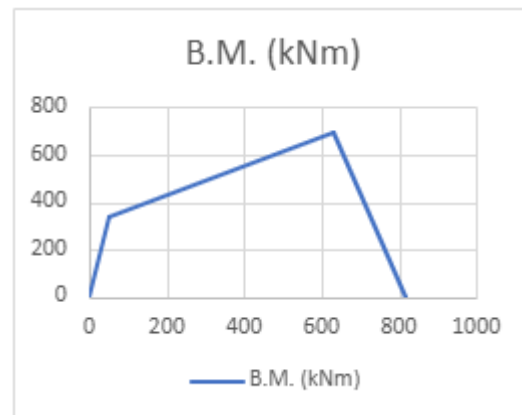
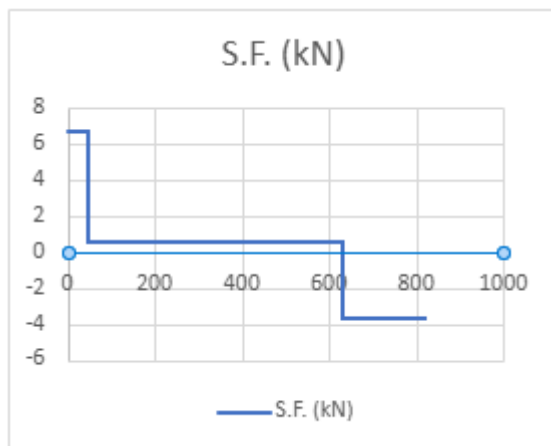
9.5.2.5.1. For radial loads

Force No.	Forces (kN)	Position (mm)	Moment around A (CC)
1	6.098887632	50	304.9443816
2	4.269221342	630	2689.609446

shaft length(mm)	Reaction B (kN)	Reaction A (kN)
820	3.651894911	6.716214063

Position (mm)	S.F. (kN)
0	6.716214063
50	6.716214063
50	0.617326431
630	0.617326431
630	-3.651894911
820	-3.651894911

Position (mm)	B.M. (kNm)
0	0
50	335.8107031
630	693.8600331
820	0



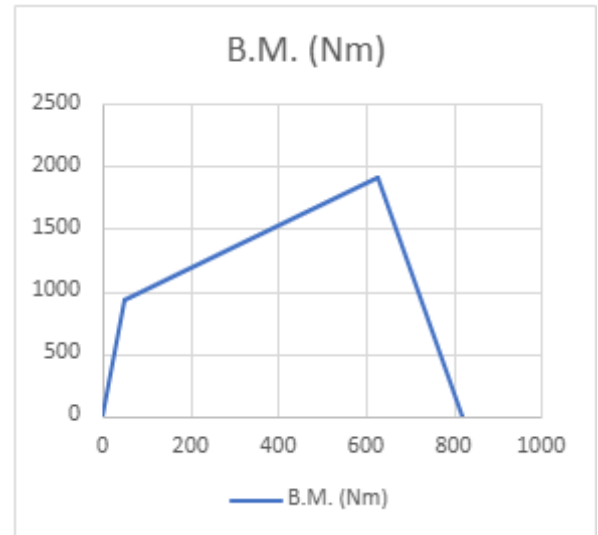
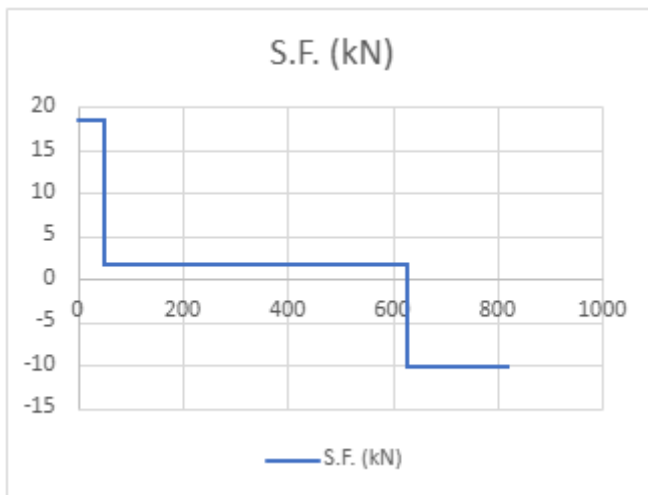
9.5.2.5.2 For tangential loads

Force No.	Forces (kN)	Position (mm)	Moment around A (CC)
1	16.75998993	50	837.9994963
2	11.73199295	630	7391.155557

shaft length(mm)	Reaction B (kN)	Reaction A (kN)
820	10.03555494	18.45642793

Position (mm)	S.F. (kN)
0	18.45642793
50	18.45642793
50	1.696438005
630	1.696438005
630	-10.03555494
820	-10.03555494

Position (mm)	B.M. (Nm)
0	0
50	922.8213965
630	1906.755439
820	0



9.5.2.6. Reverse gear

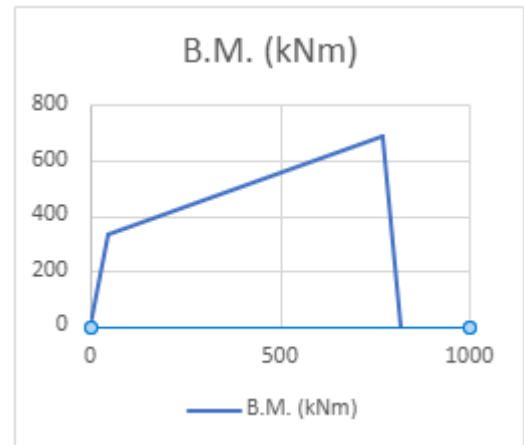
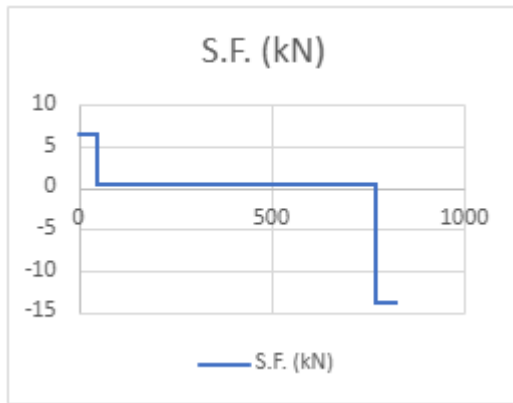
9.5.2.6.1. For radial loads

Force No.	Forces (kN)	Position (mm)	Moment around A (CC)
1	6.098887632	50	304.9443816
2	14.23073781	770	10957.66811

shaft length(mm)	Reaction B (kN)	Reaction A (kN)
820	13.73489328	6.594732155

Position (mm)	S.F. (kN)
0	6.594732155
50	6.594732155
50	0.495844523
770	0.495844523
770	-13.73489328
820	-13.73489328

Position (mm)	B.M. (kNm)
0	0
50	329.7366077
770	686.7446642
820	0



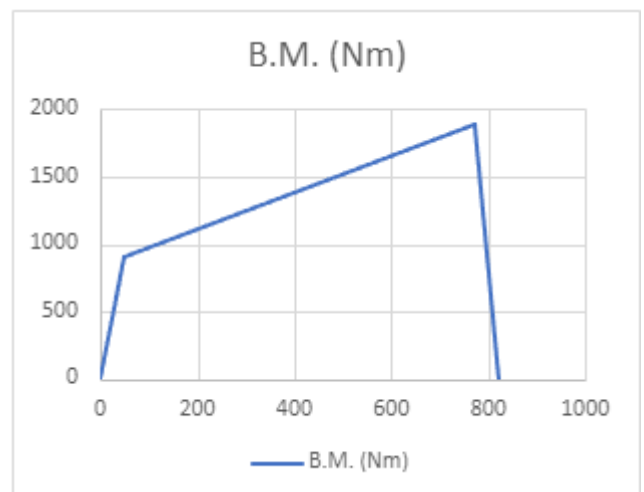
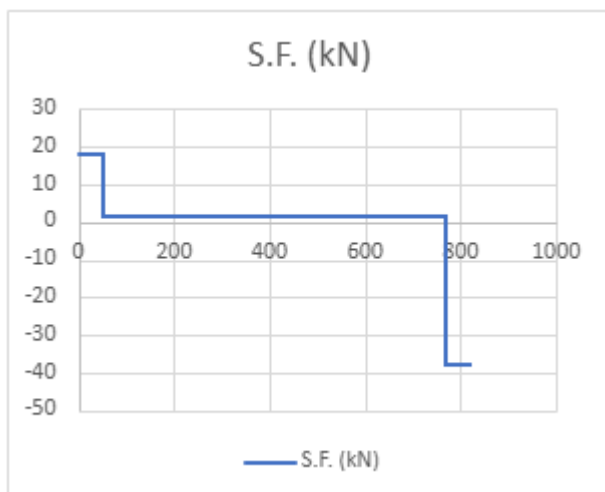
9.5.2.6.2 For tangential loads

Force No.	Forces (kN)	Position (mm)	Moment around A (CC)
1	16.75998993	50	837.9994963
2	39.10664316	770	30112.11523

shaft length(mm)	Reaction B (kN)	Reaction A (kN)
820	37.74404235	18.12259073

Position (mm)	S.F. (kN)
0	18.12259073
50	18.12259073
50	1.362600807
770	1.362600807
770	-37.74404235
820	-37.74404235

Position (mm)	B.M. (Nm)
0	0
50	906.1295366
770	1887.202118
820	0



9.6 Output shaft

- Transmitted maximum power = 172 kW
- **Final shaft diameter = 45 mm**

9.6.1 Output shaft calculation

	Transmitted Power (W) [Pt]	Shaft RPM	Safety Factor [Cs]	Design Power (W) [Pd]	Km	Kt	Transmitted Torque (Nm) [Tt]
Gear a	172000	777.777	1.5	258000	1.5	1.5	3167.641264
Gear c	172000	1102.12766	1.5	258000	1.5	1.5	2235.420277
Gear e	172000	1615.384615	1.5	258000	1.5	1.5	1525.159084
Gear g	172000	2393.548387	1.5	258000	1.5	1.5	1029.316362
Gear i	172000	3500	1.5	258000	1.5	1.5	703.9195769
Gear k	172000	777.777	1.5	258000	1.5	1.5	3167.641264

Tangential force on gear (N) [Ft]	Radial load on the tooth (N) [Fr]	Position of the gear from left side (m)	RA for Tangential force (N)	RB for Tangential force (N)	RA for Radial force (N)	RB for Radial force (N)	RA Final	RB Final
23464.00936	8538.451223	0.05	21686.43289	1777.576467	7891.598858	646.8523654	23077.66678	1891.612031
19024.85342	6923.061634	0.19	13548.00168	5476.851743	4930.059042	1993.002591	14417.13674	5828.204212
15642.65727	5692.295124	0.26	9480.398344	6162.258924	3449.875833	2242.419291	10088.58741	6557.58182
13281.50145	4833.080765	0.4	5232.106632	8049.394819	1903.940907	2929.139858	5567.758148	8565.781766
11731.99295	4269.221342	0.47	3377.391909	8354.601039	1229.018265	3040.203077	3594.059266	8890.567657
38395.65168	13972.01109	0.61	2908.761491	35486.89019	1058.485689	12913.5254	3095.365142	37763.45474
		0.66						

1st gear

Position from Left	0	0.05	0.19	0.26	0.4	0.47	0.61	0.66
Mt (Nm)		1084.321645						
Mr (Nm)		394.5799429						
Total M		1153.883339						
Me	3393.857943							
Te	5056.890878							
e	1							
tensile stress (Mpa)	864	864						
shear stress (Mpa)	432	432						
Diameter of the shaft (mm)	Under Me	Under Te	Under Me (WO keyway)	Under Te (WO keyway)				
	34.20266981	39.06519368	34.20266981	39.06519368				

2nd gear

Position from Left	0	0.05	0.19	0.26	0.4	0.47	0.61	0.66
Mt (Nm)			2574.120319					
Mr (Nm)			936.711218					
Total M			2739.25598					
Me	4706.158931							
Te	5303.433893							
e	1							
tensile stress (Mpa)	864	864						
shear stress (Mpa)	432	432						
Diameter of the shaft (mm)	Under Me	Under Te	Under Me (WO keyway)	Under Te (WO keyway)				
	38.14031869	39.69000689	38.14031869	39.69000689				

3rd gear

Position from Left	0	0.05	0.19	0.26	0.4	0.47	0.61	0.66
Mt (Nm)				2464.903569				
Mr (Nm)				896.9677166				
Total M				2623.032728				
Me	4242.929648							
Te	4551.310204							
e	1							
tensile stress (Mpa)	864	864						
shear stress (Mpa)	432	432						
Diameter of the shaft (mm)	Under Me	Under Te	Under Me (WO keyway)	Under Te (WO keyway)				
	36.84546694	37.71732928	36.84546694	37.71732928				

4th gear

Position from Left	0	0.05	0.19	0.26	0.4	0.47	0.61	0.66
Mt (Nm)					2092.842653			
Mr (Nm)					761.576363			
Total M					2227.103259			
Me	3510.424757							
Te	3680.194625							
e	1							
tensile stress (Mpa)	864	864						
shear stress (Mpa)	432	432						

Diameter of the shaft (mm)	Under Me	Under Te	Under Me (WO keyway)	Under Te (WO keyway)
	34.58984989	35.13870214	34.58984989	35.13870214

5th gear

Position from Left	0	0.05	0.19	0.26	0.4	0.47	0.61	0.66
Mt (Nm)						1587.374197		
Mr (Nm)						577.6385847		
Total M						1689.207855		
Me	2639.411208	864						
Te	2745.010634							
e	1							
tensile stress (Mpa)	864							
shear stress (Mpa)	432							
Diameter of the shaft (mm)	Under Me	Under Te	Under Me (WO keyway)	Under Te (WO keyway)				
	31.45317369	31.86716762	31.45317369	31.86716762				

Reverse gear

Position from Left	0	0.05	0.19	0.26	0.4	0.47	0.61	0.66
Mt (Nm)							1774.344509	
Mr (Nm)							645.6762702	
Total M							1888.172737	
Me	4181.906197	864						
Te	5531.553288							
e	1							
tensile stress (Mpa)	864							
shear stress (Mpa)	432							
Diameter of the shaft (mm)	Under Me	Under Te	Under Me (WO keyway)	Under Te (WO keyway)				
	36.66797135	40.25110628	36.66797135	40.25110628				

For Bearing	
RA	23077.66678
RB	37763.45474

Final diameters of the output shaft for each shear.

Maximum diameters of the output shaft under each gear (when e= 1)						
Gear	1	2	3	4	5	Reverse
Diameter of the Shaft (mm)	39.06519368	39.69000689	37.71732928	35.13870214	31.86716762	40.25110628
Final diameter of the shaft (mm)	40.25110628					
	45					

- The keyway effect is not considered for this shaft.

9.6.2. S.F. and B.M. diagrams of output shaft

9.6.2.1. First gear

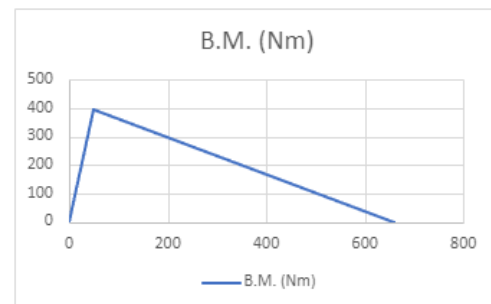
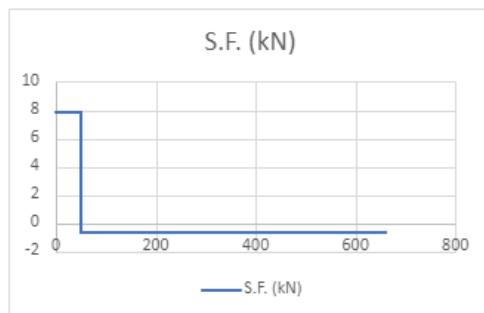
9.6.2.1.1 For radial loads

Force No.	Forces (kN)	Position (mm)	Moment around A (CC)
1	8.538451223	50	426.9225612

shaft length(mm)	Reaction B (kN)	Reaction A (kN)
660	0.646852365	7.891598858

Position (mm)	S.F. (kN)
0	7.891598858
50	7.891598858
50	-0.646852365
660	-0.646852365

Position (mm)	B.M. (Nm)
0	0
50	394.5799429
660	0



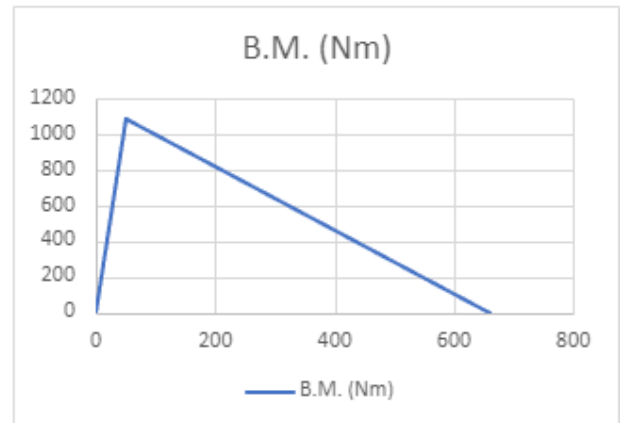
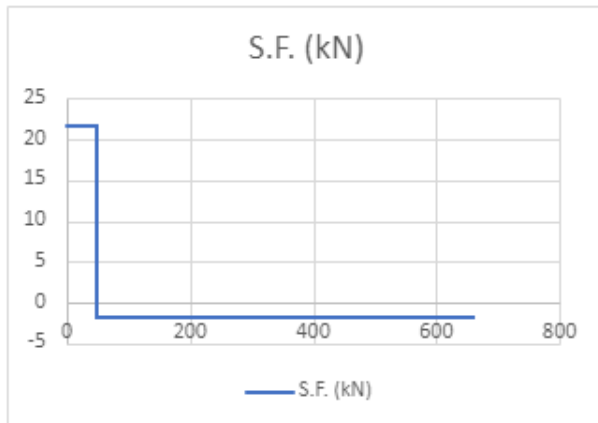
9.6.2.1.2 For tangential loads

Force No.	Forces (kN)	Position (mm)	Moment around A (CC)
1	23.46400936	50	1173.200468

shaft length(mm)	Reaction B (kN)	Reaction A (kN)
660	1.777576467	21.68643289

Position (mm)	S.F. (kN)
0	21.68643289
50	21.68643289
50	-1.777576467
660	-1.777576467

Position (mm)	B.M. (Nm)
0	0
50	1084.321645
660	0



9.6.2.2. Second gear

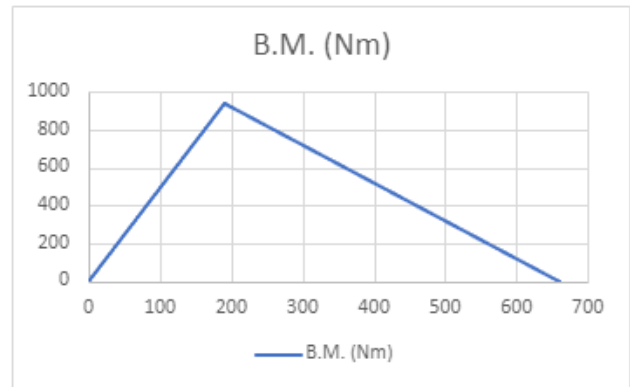
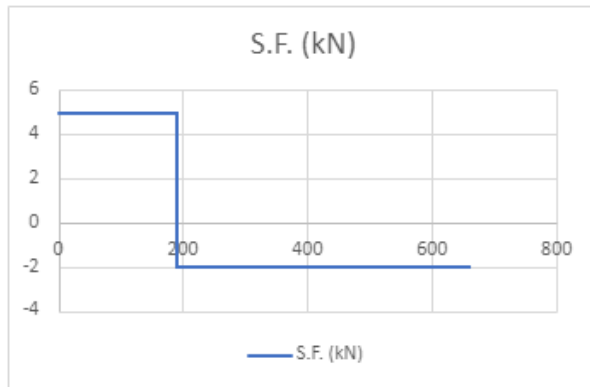
9.6.2.2.1 For radial loads

Force No.	Forces (kN)	Position (mm)	Moment around A (CC)
1	6.923061634	190	1315.38171

shaft length(mm)	Reaction B (kN)	Reaction A (kN)
660	1.993002591	4.930059042

Position (mm)	S.F. (kN)
0	4.930059042
190	4.930059042
190	-1.993002591
660	-1.993002591

Position (mm)	B.M. (Nm)
0	0
190	936.711218
660	0



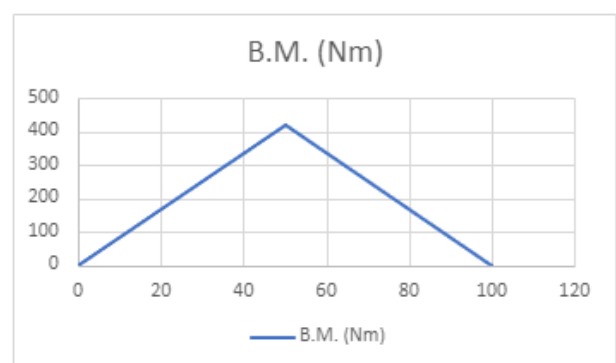
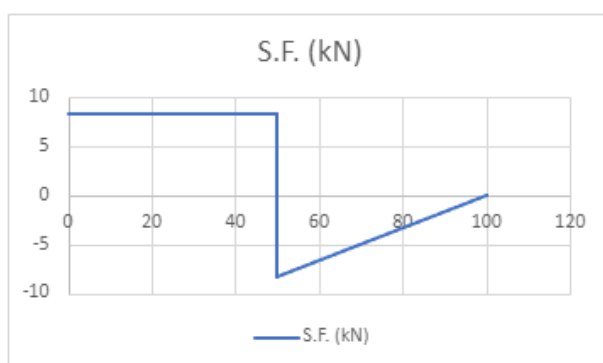
9.6.2.2.2 For tangential loads

Force No.	Forces (kN)	Position (mm)	Moment around A (CC)
1	16.75998993	50	837.9994963

shaft length(mm)	Reaction B (kN)	Reaction A (kN)
100	8.379994963	8.379994963

Position (mm)	S.F. (kN)
0	8.379994963
50	8.379994963
50	-8.379994963
100	0

Position (mm)	B.M. (Nm)
0	0
50	418.9997481
100	0



9.6.2.3. Third gear

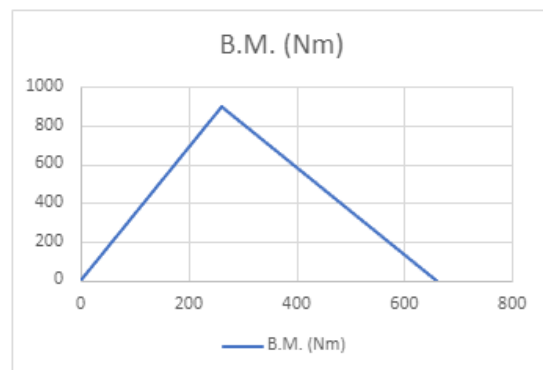
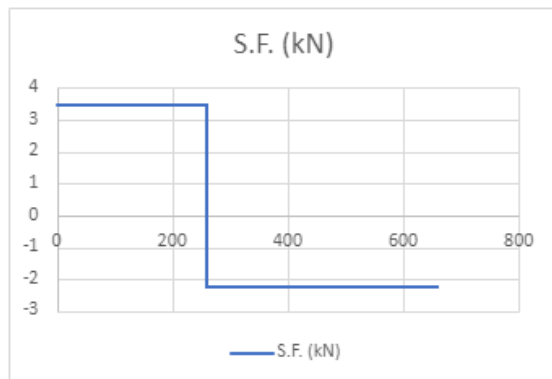
9.6.2.3.1 For radial loads

Force No.	Forces (kN)	Position (mm)	Moment around A (CC)
1	5.692295124	260	1479.996732

shaft length(mm)	Reaction B (kN)	Reaction A (kN)
660	2.242419291	3.449875833

Position (mm)	S.F. (kN)
0	3.449875833
260	3.449875833
260	-2.242419291
660	-2.242419291

Position (mm)	B.M. (Nm)
0	0
260	896.9677166
660	0



9.6.2.3.2 For tangential loads

Force No.	Forces (kN)	Position (mm)	Moment around A (CC)
1	15.64265727	260	4067.09089

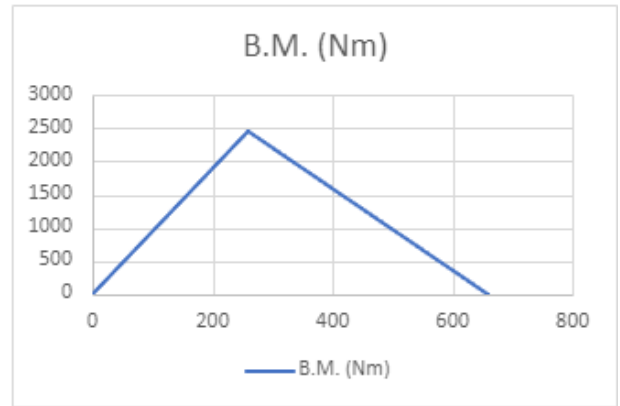
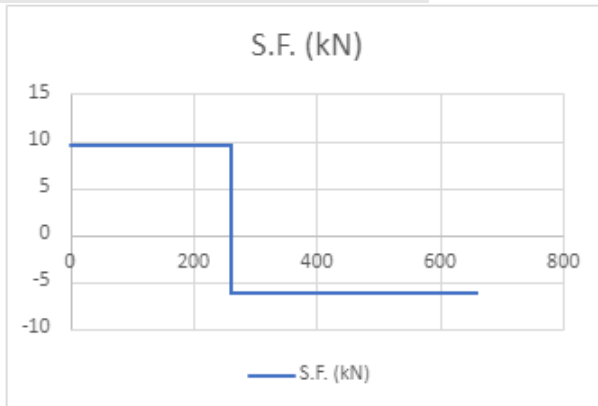
shaft length(mm)	Reaction B (kN)	Reaction A (kN)
660	6.162258924	9.480398344

Position (mm)	S.F. (kN)
0	9.480398344

Position (mm)	B.M. (Nm)
0	0

260	9.480398344
260	-6.162258924
660	-6.162258924

260	2464.903569
660	0



9.6.2.4. Fourth gear

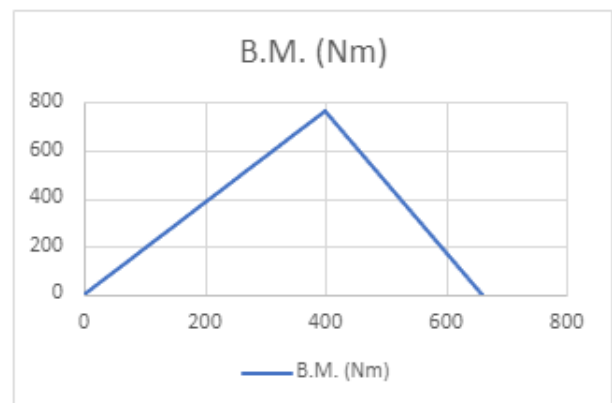
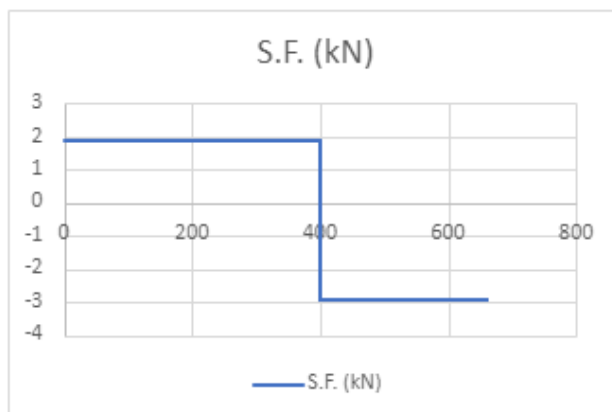
9.6.2.4.1 For radial loads

Force No.	Forces (kN)	Position (mm)	Moment around A (CC)
1	4.833080765	400	1933.232306

shaft length(mm)	Reaction B (kN)	Reaction A (kN)
660	2.929139858	1.903940907

Position (mm)	S.F. (kN)
0	1.903940907
400	1.903940907
400	-2.929139858
660	-2.929139858

Position (mm)	B.M. (Nm)
0	0
400	761.576363
660	0



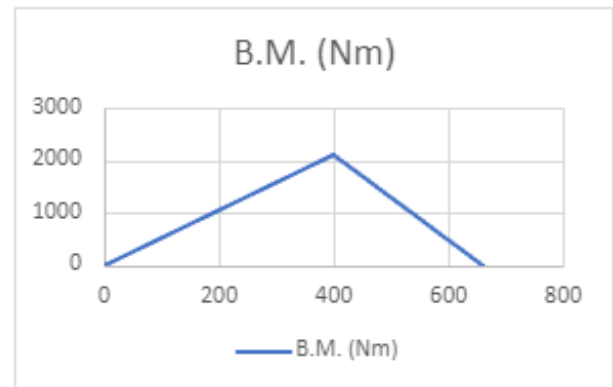
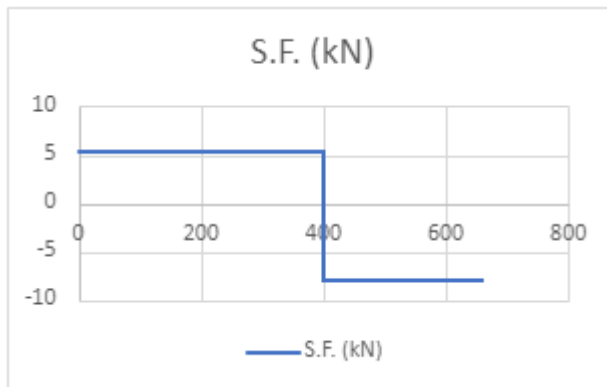
9.6.2.4.2 For tangential loads

Force No.	Forces (kN)	Position (mm)	Moment around A (CC)
1	13.28150145	400	5312.60058

shaft length(mm)	Reaction B (kN)	Reaction A (kN)
660	8.049394819	5.232106632

Position (mm)	S.F. (kN)
0	5.232106632
400	5.232106632
400	-8.049394819
660	-8.049394819

Position (mm)	B.M. (Nm)
0	0
400	2092.842653
660	0



9.6.2.5. Fifth gear

9.6.2.5.1 For radial loads

Force No.	Forces (kN)	Position (mm)	Moment around A (CC)
1	4.269221342	470	2006.534031

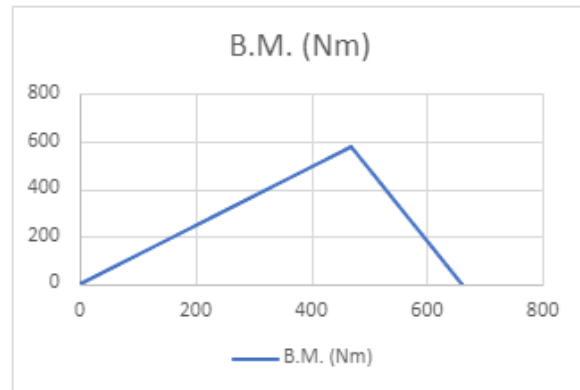
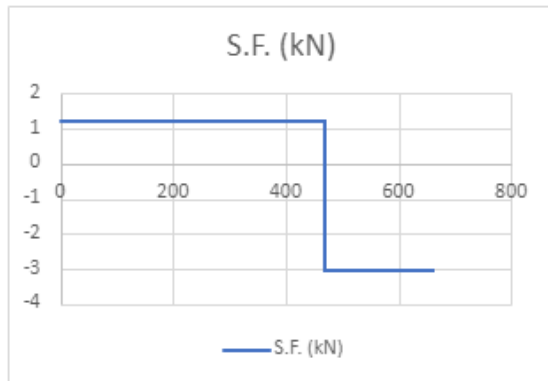
shaft length(mm)	Reaction B (kN)	Reaction A (kN)
660	3.040203077	1.229018265

Position (mm)	S.F. (kN)
0	1.229018265

Position (mm)	B.M. (Nm)
0	0

470	1.229018265
470	-3.040203077
660	-3.040203077

470	577.6385847
660	0



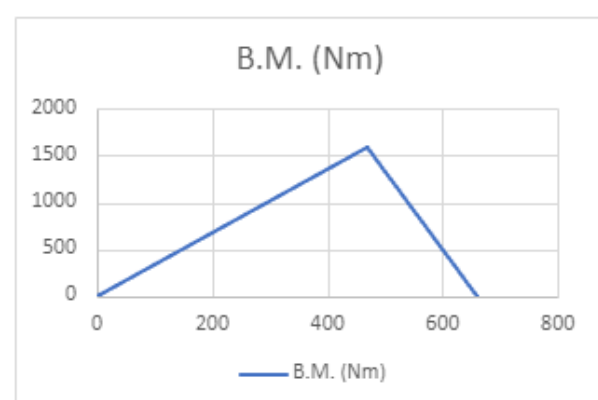
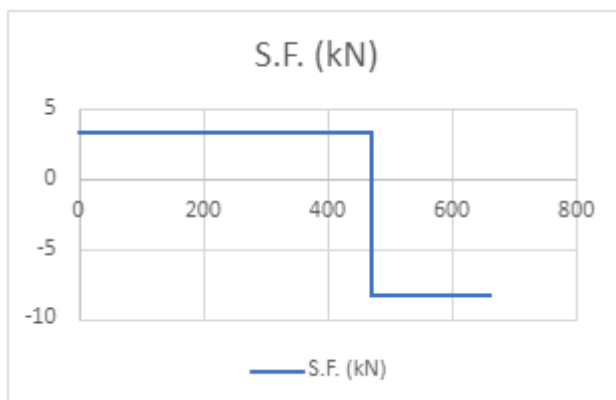
9.6.2.5.2 For tangential loads

Force No.	Forces (kN)	Position (mm)	Moment around A (CC)
1	11.73199295	470	5514.036686

shaft length(mm)	Reaction B (kN)	Reaction A (kN)
660	8.354601039	3.377391909

Position (mm)	S.F. (kN)
0	3.377391909
470	3.377391909
470	-8.354601039
660	-8.354601039

Position (mm)	B.M. (Nm)
0	0
470	1587.374197
660	0



9.6.2.6. Reverse gear

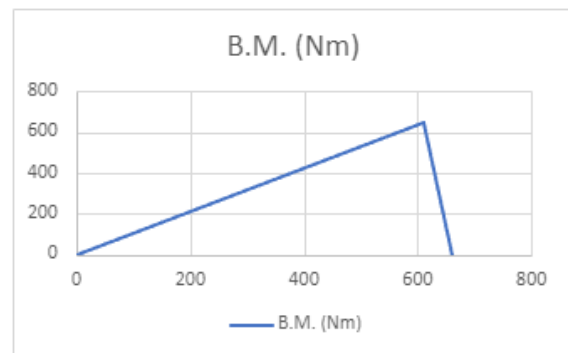
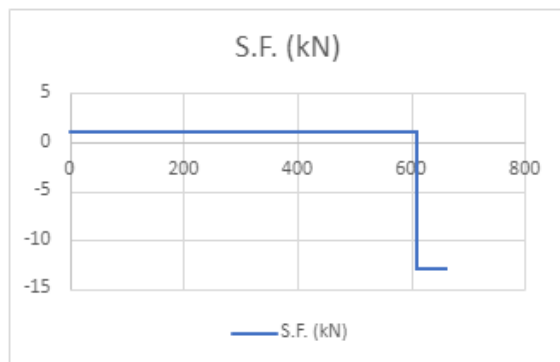
9.6.2.6.1 For radial loads

Force No.	Forces (kN)	Position (mm)	Moment around A (CC)
1	13.97201109	610	8522.926766

shaft length(mm)	Reaction B (kN)	Reaction A (kN)
660	12.9135254	1.058485689

Position (mm)	S.F. (kN)
0	1.058485689
610	1.058485689
610	-12.9135254
660	-12.9135254

Position (mm)	B.M. (Nm)
0	0
610	645.6762702
660	0



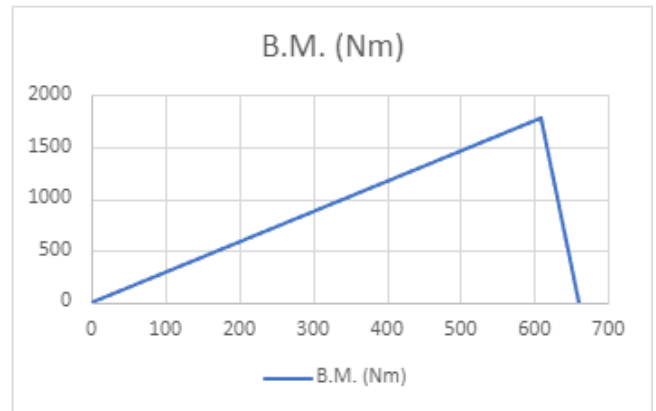
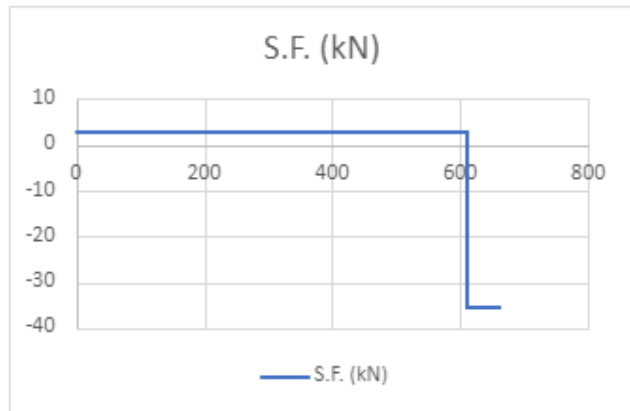
9.6.2.6.2 For tangential loads

Force No.	Forces (kN)	Position (mm)	Moment around A (CC)
1	38.39565168	610	23421.34752

shaft length(mm)	Reaction B (kN)	Reaction A (kN)
660	35.48689019	2.908761491

Position (mm)	S.F. (kN)
0	2.908761491
610	2.908761491
610	-35.48689019
660	-35.48689019

Position (mm)	B.M. (Nm)
0	0
610	1774.344509
660	0



11.Key design

We have decided to use rectangular sunk key with taper 1:100 on the top side for both idler and lay shafts since they have to transmit a higher torque. Taper effect can be neglected during calculations.

We have selected AISI 6510 steel as the key material.

Material properties

1. Yield strength $\sigma_t = 2000$ MPa
2. Factor of Safety = 1.5 (Assumed)

8.1 Keys on Lay shaft

Key parameters

For a rectangular sunk key;

$$w = \frac{d}{4} \quad t = \frac{d}{6}$$

1. Diameter of the shaft (d) = 45 mm
2. Width of the key (w) = 11.25 mm
3. Thickness (t) = 7.5 mm

Shaft diameter (mm) upto and including	Key cross-section		Shaft diameter (mm) upto and including	Key cross-section	
	Width (mm)	Thickness (mm)		Width (mm)	Thickness (mm)
6	2	2	85	25	14
8	3	3	95	28	16
10	4	4	110	32	18
12	5	5	130	36	20
17	6	6	150	40	22
22	8	7	170	45	25
30	10	8	200	50	28
38	12	8	230	56	32
44	14	9	260	63	32
50	16	10	290	70	36
58	18	11	330	80	40
65	20	12	380	90	45
75	22	14	440	100	50

Table 27 : Proportions of standard parallel, tapered and gib head keys.

However, since calculated w and t values are less than the required minimum values in the standard table, corrected values were chosen from the table.

1. $w = 16 \text{ mm}$

2. $t = 10 \text{ mm}$

Based on maximum shear stress theory,

i. Maximum allowable shear stress,

$$\tau = \frac{\sigma_{yt}}{2 \times F_s} = \frac{2000}{2 \times 1.5} = 666.67 \text{ MPa}$$

ii. Maximum allowable crushing stress,

$$\sigma_c = \frac{\sigma_{yt}}{F_s} = \frac{2000}{1.5} = 1333.33 \text{ MPa}$$

Calculations for key length

To calculate the length of the key;

i. Torsional strength of the Shaft

$$T = \frac{\pi}{16} \times \tau_{\text{shaft}} \times d^3$$

Where is the allowable shear strength of the shaft

1. Shear strength of the key

$$T = l \times w \times \tau \times \frac{d}{2}$$

- Crushing strength of the key

$$T = l \times \frac{t}{2} \times \sigma_c \times \frac{d}{2}$$

In order to transmit full power of the shaft;

Shear strength of the key = Torsional strength of the shaft

$$l \times w \times \tau \times \frac{d}{2} = \frac{\pi}{16} \times \tau_{\text{shaft}} \times d^3$$

$$l = \frac{\pi \times d^2}{8 \times w} \times \frac{\tau_{\text{shaft}}}{\tau}$$

$$= \frac{\pi \times 45^2 \times 423}{8 \times 16 \times 667} = 32.19 \text{ mm}$$

In order to key become equally strong in shearing and crushing if,

Crushing strength of the key = Shear strength of the key

$$l = \frac{\pi \times d^2}{4 \times t} \times \frac{\tau}{\sigma_c}$$

$$l = \frac{\pi \times d^2}{8 \times t} = \frac{\pi \times 45^2 \times 423}{8 \times 10 \times 1333} = 51.54 \text{ mm}$$

Considering larger value and rounding off,

$$l = 55 \text{ mm}$$

Safety check

Possible maximum torque transmitted to the key

$$T = 1759.8 \text{ Nm}$$

Therefore;

$$\tau_{\text{induced}} = \frac{T \times 2 \times 1000}{w \times l \times d} = 88.87 \text{ MPa}$$

$$\tau_{\text{induced}} < \tau_{\text{allowable}}$$

Safe for shearing

$$\sigma_c = \frac{4 \times T}{d \times t \times l}$$

$$\sigma_{c_induced} = 284.41 \text{ MPa}$$

$$\sigma_{c_induced} < \sigma_{c_allowable}$$

Safe for crushing.

Shaft strength factor, $e = 1 - 0.2 \left(\frac{w}{d} \right) - 1.1 \left(\frac{h}{d} \right)$

Where e is the shaft strength factor and h is the height of the key (t/2)

$$e = 1 - 0.2 \left(\frac{16}{45} \right) - 1.1 \left(\frac{10}{45 \times 2} \right) = 0.81$$

Therefore, strength of the shaft with the keyway

$$T_{\text{allowable}} = \frac{\pi}{16} \times \tau_{\text{shaft}} \times d^3 \times e = 6235.13 \text{ Nm}$$

$$T_{\text{allowable}} < T_{\text{max}}$$

Shaft is safe with the keyway.

In the same manner we can calculate required parameters of keys on the idler shaft.

Parameter	Lay	Idler	Input
d(mm)	45	40	32
w(mm)	16	14	12
t(mm)	10	9	8
l 1	51.54286	45.25025	32.58018
l 2	32.19014	29.06764	21.70384
l(mm)	51.54286	45.25025	32.58018
l _{selected}	55	50	35
T(Nm)	1759.8	1759.8	1759.8
τ_{in}	88.87879	125.7	261.875
$\tau_{\text{allowable}}$	667	667	667
Safety	SAFE	SAFE	SAFE
σ_{in}	303.4885	432.1155	843.9756
$\sigma_{\text{allowable}}$	1333	1333	1333
Safety	SAFE	SAFE	SAFE
e	0.806667	0.80625	0.7875
Strength	6235.127	4376.867	2188.841
Safety	SAFE	SAFE	SAFE

Table 28 : Keyway Parameters and Safety Calculations for Lay, Idler and Input Shafts

12. Bearing design

11.1. Overall procedure of bearing selection

W_R = Radial load of shaft (in N)

W_A = Axial load of shaft (in N)

L = Rating life of ball bearing (in rev)

L_H = Rating life of ball bearing (in hours)

ω = Maximum angular velocity of the shaft (in rpm)

W_{OR} = Dynamic equivalent load (in N)

$$L = 60 \times \omega \times L_H$$

X = Radial load factor

Y = Axial load factor

V = Rotation factor

$$W_{OR} = X \times V \times W_R + Y \times W_A$$

We use **double row angular contact ball bearing** for our design.

we took maximum “e” value for selecting the bearing. (According to worst case)
then, $e = 0.86$

Type of bearing	Specifications	$\frac{W_A}{W_R} \leq e$		$\frac{W_A}{W_R} > e$		e
		X	Y	X	Y	
Deep groove ball bearing	$\frac{W_A}{C_0} = 0.025$	1	0	0.56	2.0	0.22
	= 0.04				1.8	0.24
	= 0.07				1.6	0.27
	= 0.13				1.4	0.31
	= 0.25				1.2	0.37
	= 0.50				1.0	0.44
Angular contact ball bearings	Single row	1	0	0.35	0.57	1.14
	Two rows in tandem		0	0.35	0.57	1.14
	Two rows back to back		0.55	0.57	0.93	1.14
	Double row		0.73	0.62	1.17	0.86
Self-aligning bearings	Light series : for bores	1	1.3	6.5	2.0	0.50
	10 – 20 mm				2.6	0.37
	25 – 35				3.1	0.31
	40 – 45				3.5	0.28
	50 – 65				3.8	0.26
	70 – 100				3.5	0.28
	105 – 110	1	1.0	0.65	1.6	0.63
	Medium series : for bores				1.9	0.52
	12 mm				2.3	0.43
	15 – 20				2.5	0.39
	25 – 50					
	55 – 90					
Spherical roller bearings	For bores :	1	2.1	0.67	3.1	0.32
	25 – 35 mm				3.7	0.27
	40 – 45				4.4	0.23
	50 – 100				3.9	0.26
Taper roller bearings	For bores :	1	0	0.4	1.60	0.37
	30 – 40 mm				1.45	0.44
	45 – 110				1.35	0.41

Table 29 : Values of X and Y for dynamically loaded bearings (Textbook of Machine Design by Khurumi and Gupta.)

$$\frac{W_A}{W_R} = 0 ; \text{ for all situations of our design}$$

$$X = 1 \text{ and } Y = 0.73 ; \frac{W_A}{W_R} < e$$

All W_{OR} values were calculated for all bearings.

S.No.	Type of service	Service factor (K_s) for radial ball bearings
1.	Uniform and steady load	1.0
2.	Light shock load	1.5
3.	Moderate shock load	2.0
4.	Heavy shock load	2.5
5.	Extreme shock load	3.0

Table 30 : values of service factor (Textbook of Machine Design by Khurumi and Gupta.)

W = Equivalent dynamic load

k_s – service factor for ball bearing

$$W = k_s \times W_{OR}$$

C = Basic dynamic load rating

$k = 3$ for ball bearings (Textbook of Machine Design by Khurumi and Gupta.)

$$L = \left(\frac{C}{W} \right)^k \times 10^6$$

According to my C values of each bearings, bearing numbers were selected using standard table.

C_0 = Basic static load rating

All C_0 values were taken for above selected bearings using above table.

Bearing No.	Basic capacities in kN							
	Single row deep groove ball bearing		Single row angular contact ball bearing		Double row angular contact ball bearing		Self-aligning ball bearing	
	Static (C_0) (2)	Dynamic (C) (3)	Static (C_0) (4)	Dynamic (C) (5)	Static (C_0) (6)	Dynamic (C) (7)	Static (C_0) (8)	Dynamic (C) (9)
(1)								
211	26	34	30	36.5	49	53	12.7	20.8
311	42.5	56	47.5	62	80	88	28.5	58.5
411	60	78	—	—	—	—	—	—
212	32	40.5	36.5	44	63	65.5	16	26.5
312	48	64	55	71	96.5	102	33.5	68
412	67	85	—	—	—	—	—	—
213	35.5	44	43	50	69.5	69.5	20.4	34
313	55	72	63	80	112	118	39	75
413	76.5	93	—	—	—	—	—	—
214	39	48	47.5	54	71	69.5	21.6	34.5
314	63	81.5	73.5	90	129	137	45	85
414	102	112	—	—	—	—	—	—
215	42.5	52	50	56	80	76.5	22.4	34.5
315	72	90	81.5	98	140	143	52	95
415	110	120	—	—	—	—	—	—
216	45.5	57	57	63	96.5	93	25	38
316	80	96.5	91.5	106	160	163	58.5	106
416	120	127	—	—	—	—	—	—
217	55	65.5	65.5	71	100	106	30	45.5
317	88	104	102	114	180	180	62	110
417	132	134	—	—	—	—	—	—
218	63	75	76.5	83	127	118	36	55
318	98	112	114	122	—	—	69.5	118
418	146	146	—	—	—	—	—	—
219	72	85	88	95	150	137	43	65.5
319	112	120	125	132	—	—	—	—
220	81.5	96.5	93	102	160	146	51	76.5
320	132	137	153	150	—	—	—	—
221	93	104	104	110	—	—	56	85
321	143	143	166	160	—	—	—	—
222	104	112	116	120	—	—	64	98
322	166	160	193	176	—	—	—	—

Table 31 : Basic static and dynamic capacities of various types of radial ball bearings.
(Textbook of Machine Design by Khurumi and Gupta.)

$$C_0 = f_0 \times i \times Z \times D^2 \times \cos \alpha$$

f_0 = Factor depend on the type of bearing (12.3 for angular contact ball bearing) (Textbook of Machine Design by Khurumi and Gupta.)

i = No. of rows of the balls

Z = No. of balls per row

D = diameter of the balls

α = Nominal angle (20°)

In our application, there are only radial loads, so we took the smallest of the nominal angles of the standard bearings.

Diameter of the selected bearing's balls

1. Bearing 211: The ball diameter is approximately 7.938 mm. [18] [19]
2. Bearing 218: The ball diameter is approximately 9.525 mm. [20] [21]
3. Bearing 220: The ball diameter is approximately 11.112 mm. [22] [23]
4. Bearing 316: The ball diameter is approximately 15.875 mm. [22] [24]
5. Bearing 317: The ball diameter is approximately 17.462 mm. [25] [26]

Number of balls per row for selected bearing

1. Bearing 211: Typically, this bearing has 8 balls per row. [27]
2. Bearing 218: This bearing generally has 9 balls per row. [28]
3. Bearing 220: Usually, this bearing contains 10 balls per row. [29]
4. Bearing 316: This bearing often has 12 balls per row. [27]
5. Bearing 317: Typically, this bearing has 13 balls per row. [30]

When the calculated C values were compared with the C values taken from the table, it was confirmed that the relevant bearing was safe because the value applied at the relevant time was less than the maximum value in the table.

Finally, the outer diameter, inner diameter and width of the bearing corresponding to the bearing number were calculated using a standard table.

It was also concluded that there is no problem to assemble the outer diameter obtained there.

- All the equations were taken from the A Textbook of Machine Design by Khurumi and Gupta.

<i>Bearing No.</i>	<i>Bore (mm)</i>	<i>Outside diameter</i>	<i>Width (mm)</i>
200	10	30	9
300		35	11
201	12	32	10
301		37	12
202	15	35	11
302		42	13
203	17	40	12
303		47	14
403		62	17
204	20	47	14
304		52	14
404		72	19
205	25	52	15
305		62	17
405		80	21
206	30	62	16
306		72	19
406		90	23
207	35	72	17
307		80	21
407		100	25
208	40	80	18
308		90	23
408		110	27
209	45	85	19
309		100	25
409		120	29
210	50	90	20
310		110	27
410		130	31
211	55	100	21
311		120	29
411		140	33
212	60	110	22
312		130	31
412		150	35

<i>Bearing No.</i>	<i>Bore (mm)</i>	<i>Outside diameter</i>	<i>Width (mm)</i>
213	65	120	23
313		140	33
413		160	37
214	70	125	24
314		150	35
414		180	42
215	75	130	25
315		160	37
415		190	45
216	80	140	26
316		170	39
416		200	48
217	85	150	28
317		180	41
417		210	52
218	90	160	30
318		190	43
418		225	54

Table 32 : principle dimensions for radial ball bearing (Textbook of Machine Design by Khurumi and Gupta.)

11.2. Selected bearing for each shaft ending



Figure 10 : shaft

Shaft	Selected bearing number for A end	Selected bearing for B end
Input shaft	212	212
Idler shaft	218	218
Output shaft	220	317
Lay shaft	316	316

Table 31 :

bearing

final selected

11.2.1. Bearing calculation

- Basic dynamic load rating values calculation

Lifetime of the bearing			
	years	Days per 1 year	hours per day
	3	300	5
LH in h	4500		

X	1	Y	0.78	WA/WR<e	
V	1	for Angular contact double row ball bearing			
e	0.44				
Ks	2	For Moderated shock Load			
k	3	for Ball bearing		1/k	0.333333333

Shaft	Shaft Diameter (mm)	Maximum Dynamic Load on left bearing - RA (N)	Maximum Dynamic Load on right bearing - RB (N)		Shaft rpm	L in rev	Dynamic equivalent Load for RA - W_OR	Dynamic equivalent Load for RB - W_OR	Equivalent Dynamic Load - W for RA	Equivalent Dynamic Load - W for RB	Basic Dynamic Load rating for RA- C in kN	Basic Dynamic Load rating for RB- C in kN	Bearing No. for RA	Bearing No. for RB
Input	35	8917.590659	8917.590659	0	1740.95	470056500	8917.590659	8917.590659	17835.18132	17835.18132	64.36664651	64.36664651	212	212
Lay	50	35322.3591	35414.90181	0	1740.95	470056500	35322.3591	35414.90181	70644.7182	70829.80361	158.0719245	162.3204048	316	316
Output	45	23077.66678	37763.45474	0	4300	116100000	23077.66678	37763.45474	46155.33357	75526.90947	144.1050746	180.5407327	220	317
Idler	40	20807.71154	20807.71154	0	1600	432000000	20807.71154	20807.71154	41615.42308	41615.42308	117.8390134	117.8390134	218	218

- Basic static load rating values calculation

fo (depend on bearing type)	12.3	For angular contact ball bearing
i No. of rows of balls	2	
Z no. of balls per row		
Nominal angel in degrees	15	for our application

Bearing number	D_Diameters of balls in mm	Z_no. of balls per row	Basic Static Load Rating _ C0 in kN	Basic Static Load Rating in the table _ C0 in kN	safe or not
212	7.938	8	11.97818624	63	Safe
218	9.525	9	19.40221726	127	Safe
220	11.112	10	29.34021897	160	Safe
316	15.875	12	71.86006392	160	Safe
317	17.462	13	94.19117298	180	Safe

- **Final selection**

				Available serial number	
Bearing No	Bore in mm	Outer diameter in mm	Width in mm	Brand name	Serial number
212	60	110	22	SKF	3212 A
218	90	160	30	NKS	BSN4090DDUHP2BDT R BE4L5
220	90	160	30	SKF	1218
316	80	170	39	SKF	3316 A/C3
317	85	180	41	SKF	2320 KM/C3

- **Specification of the final selected bearings**

212		218			
Specifications	Details	Specifications	Details	Specifications	
Brand	SKF	Brand	NSK	Brand	
Inside Diameter	60mm	Inside Diameter	40mm	Inside Diameter	
Outside Diameter	110mm	Outside Diameter	90mm	Outside Diameter	
Race Width	36.5mm	Race Width	92mm	Race Width	
Ball Bearing Type	Double Row Angular Contact Ball Bearing	Ball Bearing Type	Single Row Angular Contact Ball Bearing	Ball Bearing Type	
Sealing Arrangement	Both Sides Sealed	Sealing Arrangement	Both Sides Sealed	Sealing Arrangement	
Bearing Clearance	CN	Bearing Clearance	CN	Bearing Clearance	
Material	Steel	Material	Steel	Material	

Ball Material	Steel	Ball Material	Steel	Ball Material	
Cage Material	Sheet Metal	Cage Material	Steel	Cage Material	
Number of Rows	2	Number of Rows	2	Number of Rows	
Static Load Rating	64kN	Static Load Rating	269kN	Static Load Rating	
Dynamic Load Rating	75kN	Dynamic Load Rating	122kN	Dynamic Load Rating	
Bore Type	Parallel	Bore Type	Parallel	Bore Type	
Bearing Series	2 Light	Partial Bearing Code	BSN4090DDUHP2BDT	Bearing Series	
Partial Bearing Code	3212	Race Type	Plain	Partial Bearing Code	
Race Type	Plain			Country of Origin	

11.3. Selected bearing for each gear on the output shaft

➤ Calculations were done as above.

- **Final selection**

Gear	Selected bearing				
	Brand name	Serial number	inner diameter (mm)	outer diameter (mm)	width (mm)
a	FAG	3209-BD-XL-TVH	45	85	60
c	FAG	4210-BB-TVH	50	65	60
e	FAG	4210-BB-TVH	50	65	60
g	FAG	4210-BB-TVH	50	65	60
i	FAG	4210-BB-TVH	50	65	60
k	SKF	7409 BCBM	45	90	60

- **Calculations**

	For WR	For WA					
Gear	Radial load on the bearing (N)	Axial load on the bearing (N) _ WA	Shaft rpm	L in rev	Dynamic equivalent Load - W_OR	Equivalent Dynamic Load _W	Basic Dynamic Load rating- C in kN
a	8538.451	0	956	258120000	8538.451	17076.90245	50.46803
c	6923.062	0	1355	365850000	6923.062	13846.12327	45.96521
e	5692.295	0	2408	650160000	5692.295	11384.59025	45.77807
g	4833.081	0	2940	793800000	4833.081	9666.16153	41.54235
i	4269.221	0	4300	1161000000	4269.221	8538.442685	41.65394
k	13972.01	0	938	253260000	13972.01	27944.02218	82.06244

- **Specification of the final selected bearings**

For gear a		For gear c		For gear e			
Brand	FAG	Brand	FAG	Brand	FAG		
Inside Diameter	45mm	Inside Diameter	50mm	Inside Diameter	50mm		Inside Diameter
Outside Diameter	85mm	Outside Diameter	65mm	Outside Diameter	65mm		Outside Diameter
Race Width	60mm	Race Width	60mm	Race Width	60mm		Race Width
Ball Bearing Type	Double Row Angular Contact Ball Bearing	Ball Bearing Type	Double Row Deep Groove Ball Bearing	Ball Bearing Type	Double Row Deep Groove Ball Bearing		Ball Bearing Type
Sealing Arrangement	Open Type	Sealing Arrangement	Open Type	Sealing Arrangement	Open Type		Sealing Arrangement
Bearing Clearance	C3	Bearing Clearance	CN	Bearing Clearance	CN		Bearing Clearance
Material	Steel	Material	Steel	Material	Steel		Material
Ball Material	Steel	Ball Material	Steel	Ball Material	Steel		Ball Material
Cage Material	Polyamide	Cage Material	Polyamide	Cage Material	Polyamide		Cage Material
Number of Rows	2	Number of Rows	2	Number of Rows		2	Number of Rows
Static Load Rating	39kN	Static Load Rating	42kN	Static Load Rating	42kN		Static Load Rating
Dynamic Load Rating	50kN	Dynamic Load Rating	46.5kN	Dynamic Load Rating	46.5kN		Dynamic Load Rating
Bore Type	Parallel	Bore Type	Parallel	Bore Type	Parallel		Bore Type
Bearing Series	2 Light	Bearing Series	2 Light	Bearing Series	2 Light		Bearing Series
Partial Bearing Code	3209	Partial Bearing Code	4210	Partial Bearing Code	4210		Partial Bearing Code
Race Type	Plain	Race Type	Plain	Race Type	Plain		Race Type
Country of Origin	Slovakia	Country of Origin	Slovakia	Country of Origin	Slovakia		Country of Origin

12. Splines on the Output shaft

12.1. Overall procedure of the spline calculation

T = Maximum torque of the shaft

n = Number of splines on the shaft

l = Spline length

D = Outer (major) diameter of the spline

d = Inner (minor) diameter of the spline

τ = Permissible shear stress

$$T = \frac{1}{8} \times \tau \times n \times l \times (D^2 - d^2)$$

We can calculate “d” by keeping “D” constant or we can calculate “D” by keeping “d” constant. If d is calculated, it must be equal to or greater than the minimum diameter of the shaft. Otherwise, the shaft will be false.

	Under constant minor diameter (d=shaft diameter)	Under constant major diameter (D=shaft diameter)
length of the hub (mm)	20	20
Number of splines	6	6
d = minor diameter of the splines (mm)	45	-
D = Major diameter of the splines (mm)	-	45
Permissible shear stress of the shaft material (Mpa)	432	432
Minimum shaft diameter (mm)	40.25110628	

	Maximum torque of the output shaft (Nm)	Dog clutch number	Selected shaft torque for design the splines	D in mm (Under constant d)	d in mm (Under constant D)	safety of the shaft when cut the sleeves on the shaft
1st gear	3167.641264	For dog clutch 1	3167.641	45.00543	44.99457	Safe
2nd gear	2235.420277					
3rd gear	1525.159084	For dog clutch 2	1525.159	50.00235	44.99738	Safe
4th gear	1029.316362					
5th gear	703.9195769	For dog clutch 3	3167.641	45.00543	44.99457	Safe
reverse gear	3167.641264					

According to the calculation the best option is to do drill the sleeve into the shaft. This minimizes material waste and reduces costs. However, it is more complex to machine and more difficult to assemble.

- But spline cutting is most practical for assembling.

W = Width of the spline

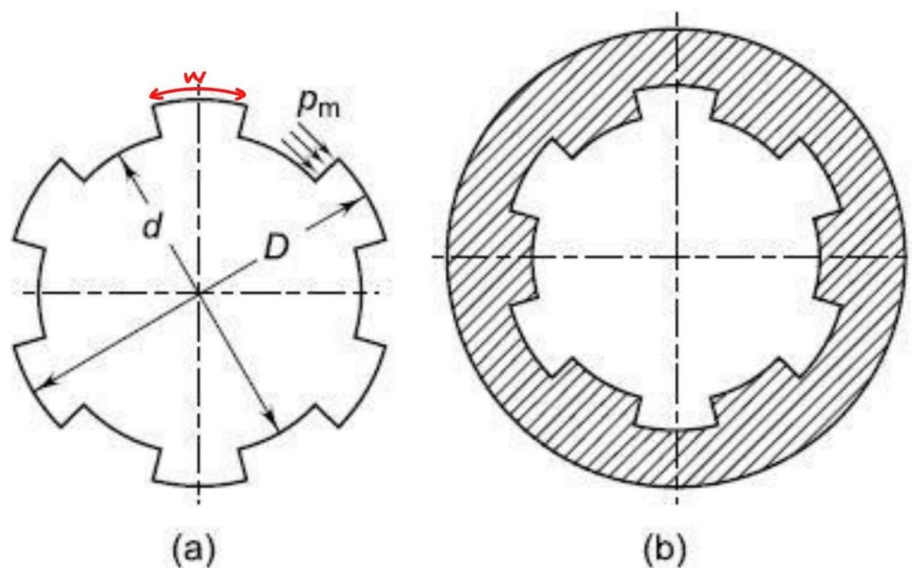
$$W = D \times 0.25$$

- All the equations were taken from the A Textbook of Machine Design by Khurumi and Gupta.

- **Final dimensions of the Splines**

For dog clutch 1 & 3		
n	6	
l	20	mm
d	45	mm
D	48	mm
w	12	mm

For dog clutch 2		
n	6	
l	20	mm
d	50	mm
D	53	mm
w	13.25	mm



Splines: (a) Shaft (b) Hub

13. Lubrication System

For a gear system operating under the conditions outlined in our design, proper lubrication is crucial for ensuring longevity, efficient operation, and minimizing wear. The lubricant must provide adequate film strength to handle the high tangential velocities and extreme forces encountered in our gearbox, especially at the maximum tangential velocity of **11.5 m/s** in the 5th gear.

13.1. Lubrication Requirements

The primary functions of the lubricant in our gearbox are:

- **Minimize friction:** Preventing metal-to-metal contact between the gear teeth to reduce wear.
- **Dissipate heat:** Managing the heat generated during high-speed operation, particularly at the maximum speed conditions.
- **Protect against corrosion and wear:** Ensuring the gear surfaces remain free from rust, oxidation, and wear due to contaminants or environmental factors.
- **Prevent foaming:** Maintaining consistent lubrication and preventing air bubbles, which could reduce the lubricant's effectiveness.

Given these operational needs, the selected lubricant must possess the following properties:

1. **Correct viscosity:** A lubricant that maintains viscosity across a wide temperature range is essential to ensure stable performance.
2. **Anti-scuffing and anti-wear properties:** To withstand the high pressures and reduce the risk of gear tooth damage.
3. **Thermal and oxidative stability:** To maintain effectiveness even at elevated temperatures and resist degradation over time.
4. **Water resistance:** Protect the gears from any moisture that could lead to rusting.
5. **Anti-foam agents:** To prevent the formation of bubbles that could impair lubrication efficiency.
6. **Anti-corrosion properties:** To protect the steel components from rust.

13.2. Lubrication Methods

Given the operational speed and load conditions of the gearbox, we evaluate three potential lubrication methods:

1. **Grease Lubrication:** This method is typically used in low-speed, light-load systems. While grease is effective in providing a protective coating, it does not offer sufficient heat dissipation for high-speed applications such as ours.
2. **Splash Lubrication:** In this method, gears are partially submerged in oil, which splashes onto them as they rotate. This technique is suitable for moderate-speed and moderate-load systems. While effective at lubricating and cooling, it might not provide the optimal protection needed for our gearbox at peak speeds.
3. **Forced Oil Circulation:** This method pumps oil under pressure directly to the gears, ensuring continuous lubrication even under high loads and speeds. The oil is circulated

through the system, allowing for better heat dissipation and providing consistent lubrication to the gears.

Considering the highest tangential velocity of **11.5 m/s** in the 5th gear, **forced oil circulation** is the most appropriate method for our gearbox. It guarantees adequate lubrication at high speeds, helps maintain the operating temperature within safe limits, and ensures durability under the heavy loads experienced in this gear.

13.3. Selection of Lubricant

For our gearbox design, we have selected a **multi-grade synthetic oil** that meets the following requirements:

- **Viscosity Grade: SAE 75W-90.** This oil provides a stable viscosity across a broad temperature range, ensuring reliable lubrication at both low and high operating temperatures.
- **Viscosity Index:** A high viscosity index (>140) ensures the lubricant maintains its protective film even during rapid gear shifts and high-speed operation.
- **Additives:** The oil contains extreme pressure (EP) additives, anti-wear agents, and anti-foaming additives to improve its performance in high-speed, high-load environments.

13.4. Maintenance and Lubrication Schedule

To maintain optimal performance and ensure long-lasting gear life, a regular oil change schedule must be followed. Given the operational demands, we recommend an oil change after every **5000 operational hours** or every **12 months**, whichever comes first. Additionally, periodic checks for oil quality, contamination, and proper levels are crucial to prevent premature wear or gear failure.

14. Design Verification and Justification

The calculated traction forces for the first and fifth gears exceed the required values. However, this provides a beneficial safety margin for the design. The following calculations verify the safety factor for each gear.

14.1. Traction Force Calculation

The traction force can be calculated using the following formula:

$$F_{traction} = \text{Torque at wheel} / \text{Tire radius}$$

Where:

Torque at wheel = Torque at output shaft $\times$ Differential ratio

Tire radius was selected as 0.38 m.

Differential ratio is given as 3.605.

First Gear

Torque at output shaft: 3167.64 Nm

Torque at wheel = 3167.64 Nm $\times$ 3.605 = 11425.91 Nm

Traction force = 11425.91 Nm / 0.38 m = 30068 N

Required traction force: 12915 N

Safety factor: 30068 N / 12915 N = 2.33

Fifth Gear

Torque at output shaft: 792 Nm

Torque at wheel = 792 Nm $\times$ 3.605 = 2855.16 Nm

Traction force = 2855.16 Nm / 0.38 m = 7513 N

Required traction force: 2313.3 N

Safety factor: 7513 N / 2313.3 N = 3.25

Justification

These safety factors are advantageous for the following reasons:

1. Handling Unexpected Loads: The increased traction allows the vehicle to perform reliably under variable conditions such as inclines, rough terrain, or when carrying heavier loads.
2. Durability: The higher forces help compensate for wear and tear, ensuring long term reliability and reducing maintenance needs.
3. Manufacturing Tolerances: The design accommodates variations in component properties, ensuring consistent performance.

In conclusion, the higher traction forces provide a necessary safety buffer, making the design suitable for real world conditions without the risk of underperformance.

15. Discussion

- **Material Selection and Adjustments**

During the material selection phase, initial choices were made based on literature review and material properties suited to each component's function. Subsequent calculations indicated that some components would benefit from alternative materials to enhance performance under load. Below are the finalized materials along with the rationale for any modifications, and suggested processes for optimal performance in real-world applications.

15.1. Gear Material: Oil-Quenched, 540°C Tempered AISI 8620

Justification – AISI 8620 was selected for its ideal case-hardening properties, providing a tough core and wear-resistant surface for gears under high-contact stresses. Calculation results confirmed its suitability, so no material change was necessary.

Suggested Processing

- **Heat Treatment:** Oil quenching and tempering at 540°C is recommended to achieve a hard, wear-resistant surface while maintaining core ductility. This treatment balances strength and toughness, enhancing gear durability under cyclic loads.
- **Machining:** Precision grinding and milling are advisable to achieve the required tooth profiles and surface finish for optimal performance.

15.2. Shaft Material: AISI 4340 (Replacing AISI 1045)

Justification – AISI 4340 was chosen over AISI 1045 to provide the necessary increase in tensile strength and fatigue resistance, as calculations showed AISI 1045 might fail under operational stresses. AISI 4340 offers a higher strength-to-weight ratio, critical for withstanding torsional and bending loads in the shaft.

Suggested Processing

- **Heat Treatment:** Quenching and tempering in the range of 315–425°C would improve the tensile strength and toughness of AISI 4340, optimizing it for dynamic loads.
- **Machining:** Standard turning and grinding are recommended to achieve precise tolerances and smooth surfaces for assembly alignment.

15.3. Bearing Material: SAE 52100 Standard Bearing Steel

Justification – SAE 52100 was chosen initially for its high hardness and wear resistance, crucial for enduring rolling and fatigue stresses. This material proved suitable during calculations, so no changes were required.

Suggested Processing

- **Heat Treatment:** To maximize surface hardness, controlled heating and quenching are suggested, achieving hardness around 60–67 HRC, essential for reducing wear in bearing applications.
- **Machining:** Precision grinding is advised due to the material's high hardness post-treatment, allowing for dimensional accuracy and a smooth finish to minimize friction.

15.4. Key Material: AISI 6510 (Replacing AISI 1045)

Justification – AISI 6510 was selected over AISI 1045 for its superior yield strength, necessary to resist shear forces within the keyway. Calculations indicated that AISI 1045 might lack sufficient yield strength, whereas AISI 6510 provides better reliability under load.

Suggested Processing

- **Heat Treatment:** Quenching followed by tempering around 250–350°C would enhance the material's yield strength, making it more resilient to shear.
- **Machining:** Shaping and finishing via grinding or milling are advised to ensure accurate fitment and reliable force transmission in the keyway.

16. References

- [1] “2024 Toyota Highlander | Toyota.com.” Accessed: Sep. 08, 2024. [Online]. Available: <https://www.toyota.com/highlander/>
- [2] “210321M-Highlander-Tech-Spec.pdf.” Accessed: Sep. 08, 2024. [Online]. Available: <https://media.toyota.co.uk/wp-content/uploads/sites/5/pdf/210321M-Highlander-Tech-Spec.pdf>
- [3] “Chevrolet Camaro Z28 SS 5.7 V8 - [2002] Performance Figures, Specs and Road Legal Technical Information.” Accessed: Sep. 08, 2024. [Online]. Available: <https://www.autosnout.com/Figures-Specs/Chevrolet-Camaro-Z28-SS-5-7-V8-2002/1265>
- [4] “How Much Does the Average Man Weigh?,” Healthline. Accessed: Sep. 09, 2024. [Online]. Available: <https://www.healthline.com/health/mens-health/average-weight-for-men>
- [5] “Tire Selector.” Accessed: Sep. 11, 2024. [Online]. Available: https://www.activegreenross.com/tire-selector?tires=&altsizevalue=&filter_order=&type=&searchvehicle=
- [6] “How To Read A Tyre | Car maintenance, Car mechanic, Car gadgets,” Pinterest. Accessed: Sep. 08, 2024. [Online]. Available: <https://www.pinterest.com/pin/how-to-read-a-tyre--576953402293235016/>
- [7] “(5) Video | Facebook.” Accessed: Sep. 10, 2024. [Online]. Available: <https://www.facebook.com/watch/?v=1222483158151458>
- [8] S. K. City, “Sporting KC Arts: Color your Toyota Highlander,” Sporting Kansas City. Accessed: Sep. 10, 2024. [Online]. Available: <https://www.sportingkc.com/news/sporting-kc-arts-color-your-toyota-highlander>
- [9] “2018-2019年丰田汉兰达维修手册电路图资料下载_汽修帮手_让汽修更简单_汽车维修手册电路图下载_维修资料库.” Accessed: Sep. 10, 2024. [Online]. Available: <https://www.jizwx.com/hanxiche/1745.html>
- [10] “(PDF) Understanding Air Pollution Dynamics and Exploring Reasons for Aggravation of Air Pollution in the City of Colombo Using Geo-Statistical Techniques.” Accessed: Sep. 10, 2024. [Online]. Available:

https://www.researchgate.net/publication/272686901_Understanding_Air_Pollution_Dynamics_and_Exploring_Reasons_for_Aggravation_of_Air_Pollution_in_the_City_of_Colombo_Using_Geo-Statistical_Techniques

[11] “Faizan Nawaz 2222.pdf.” Accessed: Sep. 10, 2024. [Online]. Available: https://www.automechanikadubai.com/uploads/editor_images/file/Automechanika17ppts/Faizan%20Nawaz%202222.pdf

[12] Accessed: Sep. 10, 2024. [Online]. Available: http://www.thecartech.com/KnowYourCar/Transmission/Transmission_system.htm

[13] rotontek, “What is the difference between drive shaft and drive axle?,” Rotontek. Accessed: Sep. 10, 2024. [Online]. Available: <https://rotontek.com/what-is-the-difference-between-drive-shaft-and-drive-axle/>

[14] x-engineer.org, “Drivetrain losses (efficiency) – x-engineer.org.” Accessed: Sep. 10, 2024. [Online]. Available: <https://x-engineer.org/drivetrain-losses-efficiency/>

[15] “For those had/have torque converter shudders and judders, what do they feel like? | Toyota Nation Forum.” Accessed: Sep. 10, 2024. [Online]. Available: <https://www.toyotanation.com/threads/for-those-had-have-torque-converter-shudders-and-judders-what-do-they-feel-like.1036417/>

[16] “L375-30 Automotive Engine | Dongfeng Cummins-DCEC.” Accessed: Sep. 10, 2024. [Online]. Available: <https://www.dcec-engine.com/dcec-l375-30-automotive-engine/>

[17] “Cummins L375-30 Automotive Engine,” RemiPower. Accessed: Sep. 10, 2024. [Online]. Available: <https://remipower.com/>

[18] NTN Corporation, “Technical Data,” Accessed: Sep. 10, 2024. [Online]. Available: https://www.ntnglobal.com/en/products/catalog/pdf/2250E_03.pdf

[19] SKF, “QJ-211-MA Angular Contact Ball Bearing,” RS Online, Accessed: Sep. 10, 2024. [Online]. Available: <https://us.rs-online.com/product/skf/qj-211-ma/73221007/>

[20] SKF, “Four-Point Contact Ball Bearing QJ 218 N2MA,” Accessed: Sep. 10, 2024. [Online]. Available: <https://www.skf.com/group/products/rolling-bearings/ball-bearings/angular-contact-ball-bearings/four-point-contact-ball-bearings/productid-QJ%2018%20N2MA>

- [21] SKF, "QJ-218-N2MA Angular Contact Ball Bearing," RS Online, Accessed: Sep. 10, 2024. [Online]. Available: <https://us.rs-online.com/product/skf/qj-218-n2ma/73221017/>
- [22] Timken, "Angular Contact Ball Bearings," Accessed: Sep. 10, 2024. [Online]. Available: <https://www.timken.com/products/timken-engineered-bearings/ball/angular-contact/>
- [23] NTN Corporation, "Technical Data," Accessed: Sep. 10, 2024. [Online]. Available: https://www.ntnglobal.com/en/products/catalog/pdf/2203E_b04.pdf
- [24] SKF, "QJ-316-N2MA Angular Contact Ball Bearing," RS Online, Accessed: Sep. 10, 2024. [Online]. Available: <https://us.rs-online.com/product/skf/qj-316-n2ma/73221039/>
- [25] SKF, "QJ-317-N2MA Angular Contact Ball Bearing," RS Online, Accessed: Sep. 10, 2024. [Online]. Available: <https://us.rs-online.com/product/skf/qj-317-n2ma/73221042/>
- [26] SKF, "Double-Row Angular Contact Ball Bearing 3217 A," Accessed: Sep. 10, 2024. [Online]. Available: <https://www.skf.com/ca/en/products/rolling-bearings/ball-bearings/angular-contact-ball-bearings/double-row-angular-contact-ball-bearings/productid-3217%20A>
- [27] NTN Corporation, "Technical Data," Accessed: Sep. 10, 2024. [Online]. Available: https://www.ntnglobal.com/en/products/catalog/pdf/2250E_03.pdf
- [28] SKF, "QJ-218-N2MA Angular Contact Ball Bearing," RS Online, Accessed: Sep. 10, 2024. [Online]. Available: <https://us.rs-online.com/product/skf/qj-218-n2ma/73221017/>
- [29] Schaeffler Group, "Technical Product Information: Bearings for Screw Drives," Accessed: Sep. 10, 2024. [Online]. Available: https://www.schaeffler.com/remotemedien/media/_shared_media/08_media_library/01_publications/schaeffler_2/tpi/downloads_8/tpi_213_de_en.pdf
- [30] SKF, "QJ-317-N2MA Angular Contact Ball Bearing," RS Online, Accessed: Sep. 10, 2024. [Online]. Available: <https://us.rs-online.com/product/skf/qj-317-n2ma/73221042/>